

Solana and the SOL Market

Craig M. Lewis and James. A. Overdahl¹

January 29, 2025

¹ Craig Lewis is a senior advisor at Patomak Global Partners and the Madison S. Wigginton Professor of Finance at the Owen Graduate School of Management at Vanderbilt University. James Overdahl is a partner at Delta Strategy Group and a specialist in the regulation of financial and commodity markets. The authors are appreciative of the following organizations who provided support and thoughtful feedback on this paper: Solana Foundation, Anagram Ltd., Anza Technology, Inc., Block Logic, LLC, DoubleZero Foundation, Helius Blockchain Technologies, Inc., Jito Labs, Multicoin Capital, Sol Strategies Inc., and Sphere Laboratories, Inc.

Table of Contents

I.	INTRODUCTION	1
II.	DIGITAL ASSETS, BLOCKCHAINS, AND THE SOLANA DECENTRALIZED NETWORK.....	2
A.	Blockchains.....	3
1.	Bitcoin.....	4
2.	Ethereum.....	6
B.	The Solana Decentralized Network	11
1.	How transactions work	13
2.	Proof of History (PoH) and Consensus.....	15
3.	Economic model	16
4.	Decentralization	17
III.	THE SOL TOKEN AND MARKET METRICS.....	19
A.	Tokenomics.....	20
B.	SOL market structure.....	23
1.	CEX Volumes	24
2.	DEX Volumes.....	27
C.	Examining the SOL market: measures of liquidity, market quality, and susceptibility to manipulation.....	32
1.	Liquidity and Market Quality Measures	34
2.	SOL returns are highly correlated across exchanges	51

I. INTRODUCTION

1. Institutional investors' growing interest in cryptocurrencies marks a significant milestone for digital asset adoption. Bitcoin, in particular, has become a focal point because of its established position in the broader digital asset landscape.² Meanwhile, Ethereum, with its robust smart contract capabilities and sizable decentralized finance (DeFi) ecosystem, has also attracted institutional attention.³ This heightened interest in both assets is illustrated by the availability of regulated exchange-traded products and futures contracts, which allow investors to gain exposure without directly holding the underlying cryptocurrencies.⁴

2. While Bitcoin and Ethereum have been at the forefront of the public discussion of cryptocurrencies due to their sizable spot markets and institutional interest, they are by no means the only blockchain networks that have attracted investor attention. The market for digital assets now includes additional multi-purpose network blockchains with the potential to exceed limitations found in early systems, most notably Bitcoin and Ethereum.

3. In this paper, we aim to provide a high-level yet detailed review of Solana and the market for its native token, SOL, and we provide context for evaluating the SOL market relative to the markets for BTC and ETH – two markets for which regulated institutional finance products have been approved. We separate the discussion into two parts. Section II provides an overview of digital assets and blockchains, and a detailed review of the Solana decentralized network along with its key technological advantages. Section III discusses the SOL token,⁵ the cryptocurrency associated with the Solana blockchain and the market in which it trades. Section III also provides a detailed review of the SOL market structure.

² See e.g., “North America: Institutional Momentum and U.S. Bitcoin ETPs Propel Crypto Further Into the Mainstream,” Chainalysis, October 17, 2024, available at <https://www.chainalysis.com/blog/north-america-crypto-adoption-2024/>.

³ See e.g., “Staying the course: institutional investor outlook on digital assets,” EY-Parthenon, May 10, 2023, available at https://www.ey.com/en_us/insights/financial-services/how-institutions-are-investing-in-digital-assets.

⁴ Spot Bitcoin Exchange-Traded Products were approved in January 2024. See “Statement on the Approval of Spot Bitcoin Exchange-Traded Products,” Chair Gary Gensler, January 10, 2024, available at <https://www.sec.gov/newsroom/speeches-statements/gensler-statement-spot-bitcoin-011023>; Spot ETH Exchange-Traded Products were approved in July 2024. See e.g., Braun, Helene, “Ethereum ETFs Approved by SEC, Bringing Popular Funds to Second-Largest Cryptocurrency,” CoinDesk, July 22, 2024, available at <https://www.coindesk.com/business/2024/07/22/sec-approves-spot-ethereum-etfs>.

⁵ Throughout this document, we refer to SOL as a “token” rather than a “coin” for consistency and simplicity, even though SOL is technically a coin, as it serves as the native cryptocurrency of the Solana blockchain. See Section III.A.

II. DIGITAL ASSETS, BLOCKCHAINS, AND THE SOLANA DECENTRALIZED NETWORK

4. The term “digital assets” has been applied broadly to describe anything of value that is subject to digital storage, such as information recorded in digital formats. Such a definition encompasses digitally formatted databases, software, documents, books, websites, and video content. These digital assets have property attributes representing claims of ownership and can therefore be bought, sold, traded, or held as a store of value.

5. In recent years, the term “digital assets” has evolved to include digital (or virtual) currencies. A digital currency is any currency that is stored or exchanged on a network of digital computers. Types of digital currencies include cryptocurrency and central bank digital currency (“CBDC”).⁶ Digital currency may be recorded through a decentralized network of users or on a centralized ledger operated by a financial institution. Digital currencies exhibit properties similar to traditional currencies but generally do not have a physical form like printed banknotes or minted coins. Instead, their digital form allows for nearly instantaneous transactions over the internet. Digital currencies allow for peer-to-peer transactions, including transactions internationally across borders.

6. Digital assets include cryptocurrencies like Bitcoin, ETH, and SOL and other tokenized digital assets, such as utility or governance tokens, non-fungible tokens (“NFTs”), gaming or data tokens, or tokenized real estate. These types of cryptocurrencies may be used to buy physical goods and services but may also be restricted to certain communities.

7. This paper focuses on cryptocurrencies, particularly SOL. Below, we begin briefly with a high-level overview of the Bitcoin and Ethereum blockchains, as they contain characteristics that aid in understanding Solana and also offer relevant comparisons for the market structure discussion in Section III. We then provide a more detailed discussion of the Solana blockchain and its key technological advantages.

⁶ For a discussion of the potential uses for CBDCs, *see* “Central bank digital currencies,” Committee on Payments and Market Infrastructures of Bank for International Settlement (BIS), March 2018, available at <https://www.bis.org/cpmi/publ/d174.pdf>.

A. Blockchains

8. A blockchain refers to a decentralized digital ledger technology that facilitates secure, immutable, and transparent transactions without a central intermediary. Every transaction on a blockchain is cryptographically validated and constitutes an “update” to the ledger. These updates are organized into “blocks,” which is a way to batch them for the network to agree on its current state. Blocks are linked to form an “update chain,” which makes the current state easily verifiable and its history transparent. Together, the blocks that form the “chain” result in a ledger that is shared, replicated, and synchronized among the members of a decentralized network.⁷ Each user of the ledger could have one or more public blockchain addresses with a corresponding private key for each address.⁸ A private key differs from a traditional password in that it is not kept by a central authority that determines whether the user has the correct credentials to use the address. Instead, the blockchain relies upon cryptography to digitally authenticate ownership of the public key without revealing the private key. The cryptocurrency associated with each address can be spent only by the person with the private key credential corresponding to the address.⁹

9. Within a blockchain network, there are (i) a common set of rules that determine which transactions are valid and how the network is updated (“validity rules”¹⁰) and (ii) a “consensus mechanism” that ensures security, maintains a consistent ledger, and processes and validates transactions.¹¹ These mechanisms facilitate agreement among the participants of the distributed network on the integrity of the ledger and prevent the same token from being spent twice (“double-spending”). Collectively, the cryptographic, consensus, and validity rules that govern a

⁷ See e.g., Garrick Hileman and Michel Rauchs, “2017 Global Blockchain Benchmarking Study,” (2017) (“Hileman and Rauchs (2017)”).

⁸ Wallet software cryptographically generates user keypairs, including both a public key and a private key. Public keys are known to all participants on the network and provide a unique identifier for economic activities (e.g., interacting with DeFi applications). Whereas public keys are known and seen by all, private keys are not.

⁹ Passkeys employed by single sign-on (SSO) apps and platforms, such as Apple iOS, leverage the same type of asymmetric cryptography. As explained above, in this approach, users authenticate using their unique identifier and private key, effectively replacing traditional username/password combinations with a different method of identity verification. See e.g., Annie Badman and Matt Kosinski, “What is asymmetric encryption?,” IBM, August 2024, available at <https://www.ibm.com/think/topics/asymmetric-encryption>; See also, “Public Vs Private Keys: Understanding PKC in Crypto” Ledger Academy, October 23, 2019, available at <https://www.ledger.com/academy/blockchain/what-are-public-keys-and-private-keys>.

¹⁰ Hileman and Rauchs (2017), p. 14.

¹¹ Hileman and Rauchs (2017), p. 14. See also, “Consensus Mechanism,” Ledger Academy, September 13, 2023, available at <https://www.ledger.com/academy/consensus-protocols-how-are-blockchains-secure>.

blockchain's operations are referred to as its "protocol," which can be quite different across different blockchains.¹²

1. Bitcoin

10. Because of its position as the largest cryptocurrency by market capitalization,¹³ the development of Bitcoin in 2008-2009 is a useful reference point for understanding the scope, context, and features of digital assets like SOL. Bitcoin traces its origins to a nine-page white paper that was published in 2008 under the pseudonym of Satoshi Nakamoto, whose identity remains unknown.¹⁴ Bitcoin enables peer-to-peer transactions without the involvement of a trusted third party. It functions as a self-sustaining digital currency that does not rely on a central bank or financial institution. Instead, Bitcoin operates through a system in which every peer-to-peer transaction and each user's balance are tracked and recorded on a community-managed public ledger known as the blockchain. Notably, although the words "block" and "chain" appear separately in the original 2008 Bitcoin white paper, the combined term "blockchain" does not. Unlike a traditional private ledger, a blockchain ledger is stored by each participant in the network.

11. Bitcoin's blockchain functions as a shared, public ledger that records all confirmed transactions in sequential blocks. Each block contains a list of newly broadcasted transactions and a reference to the previous block, creating an unbroken chain that can be traced back to the very first block. Bitcoin uses a Proof of Work ("PoW") consensus mechanism to maintain the blockchain's security and integrity. This process relies on a specific type of mathematical operation called hashing.

12. Hashing on Bitcoin's blockchain is performed by running a data set—such as a block's transactions, its timestamp, and a reference to the most recent block—through a cryptographic function called SHA 256.¹⁵ This function produces a fixed-length "hash," commonly shown as a

¹² See e.g., "Consensus Mechanisms," Ethereum.org, January 13, 2023, available at <https://ethereum.org/en/developers/docs/consensus-mechanisms/>; See also, "What is a Protocol," Coinbase, available at <https://www.coinbase.com/en-gb/learn/crypto-basics/what-is-a-protocol>.

¹³ See "Today's Cryptocurrency Prices by Market Cap," CoinMarketCap, last accessed January 17, 2025, available at <https://coinmarketcap.com/>.

¹⁴ Satoshi Nakamoto, "Bitcoin: Peer-to-Peer Electronic Cash System," White Paper, Cryptography Correspondence Group, October 31, 2008 ("Nakamoto (2008)"). Available at: <https://bitcoin.org/bitcoin.pdf>.

¹⁵ Nakamoto (2008), p. 3.

64-character sequence of letters and numbers. A key property of this function is that even the slightest change to the input data completely alters the resulting hash. This makes the hash a unique digital fingerprint of the data at that point in time.¹⁶

13. Transactions submitted by users but not yet processed by the network are held in a memory pool (“mempool”).¹⁷ As an example, suppose a user wishes to send BTC from their wallet to another user’s. After initiating the transaction, it is broadcast to the Bitcoin network and temporarily held in the mempool. Miners, who are the participants responsible for adding new blocks, gather unconfirmed transactions, confirm the transaction is valid (e.g., the user has the BTC they wish to send), and then package them into a candidate block.¹⁸

14. Next, they must find a hash that meets a specific requirement known as the “target.”¹⁹ The target is a threshold number, and the block’s hash must be lower than this target. Because the output of the hash function is unpredictable, miners must repeatedly try different inputs to achieve the desired result. They do this by altering a small portion of the block’s data called the “nonce.”²⁰ By changing the nonce and re-hashing, they run through a trial-and-error process until they find a hash that falls below the target. This process is computationally intensive and expensive, ensuring that producing a valid block requires significant work.

15. This difficulty confers robust security benefits. If someone attempts to modify a previously confirmed transaction, the block’s hash will no longer match what the network accepted. To restore consistency, the attacker would need to redo the Proof of Work not just for that block, but for every block that follows. As the chain grows, this quickly becomes nearly impossible. Thus, the cost and effort to rewrite history is prohibitively high, making the blockchain tamper-resistant.

16. Additionally, the hashing and PoW processes help prevent double spending. Any attempt to insert a conflicting transaction later would fail because each new block depends on the exact

¹⁶ See, e.g., Upadhyay et al., “Investigating the Avalanche Effect of Various Cryptographically Secure Hash Functions and Hash-Based Applications,” *IEEE Access*, vol. 10, pp. 112472-112486, 2022.

¹⁷ See e.g., “What Is a Mempool?” Ledger Academy, May 1, 2023, available at <https://www.ledger.com/academy/what-is-a-mempool>.

¹⁸ See e.g., “Confirmed and unconfirmed blockchain transactions,” Bitstamp Learn, August 5, 2022, available at <https://www.bitstamp.net/learn/blockchain/confirmed-and-unconfirmed-blockchain-transactions/>.

¹⁹ See e.g., “Bitcoin Mining Difficulty: Everything You Need to Know,” Coindesk, March 18, 2022, available at <https://www.coindesk.com/learn/bitcoin-mining-difficulty-everything-you-need-to-know>.

²⁰ See e.g., “Nonce,” Binance Academy, available at <https://academy.binance.com/en/glossary/nonce>.

sequence and state of all prior blocks. The block's hash would not match what the rest of the network expects, and it would be rejected.

17. The consensus among users therefore results from a cryptographic PoW function and miners are rewarded with Bitcoins as an incentive to run the transaction network. The probability of receiving a reward is proportional to the PoW, that is, the computational effort expended. With a PoW mechanism, miners compete to find a valid hash for a new block. The “winning” miner is the first to solve the puzzle and is rewarded with transaction fees and newly created Bitcoins (the “mining award” or “coinbase” reward).²¹ Although PoW provides strong security guarantees by making it economically impractical for a single entity to alter the ledger, it faces limitations in scalability and energy efficiency due to the resource-intensive nature of mining.

18. The rate at which Bitcoin adds new blocks, referred to as “block time,” is regulated by the difficulty adjustment mechanism, which ensures that blocks are mined approximately every 10 minutes.²² Currently, new blocks are added approximately every 10 minutes. The ten-minute block time offers a balance between transaction speed and network security.

19. While Bitcoin and its PoW consensus provide a relatively robust and secure network conducive to a single digital currency, any further applications are limited by its long block time and computationally intensive consensus mechanism. The Ethereum protocol addresses these issues.

2. Ethereum

20. Like Bitcoin, Ethereum is a decentralized blockchain that relies on a network of nodes to reach consensus and record transactions.²³ Unlike Bitcoin, the Ethereum blockchain was not created primarily to host a digital currency. As a consequence, Ethereum's protocol differs in important ways from that of Bitcoin.

21. Ethereum's main innovation lies in its ability to run self-executing contracts with predefined rules (“smart contracts”), enabling the creation and operation of decentralized

²¹ Note that this technical terminology is unrelated to Coinbase Global, Inc.

²² The network adjusts the difficulty of mining approximately every 2,016 blocks in order to maintain the 10 minute target. See e.g., “Bitcoin Mining Difficulty: Everything You Need to Know,” Coindesk, March 18, 2022, available at <https://www.coindesk.com/learn/bitcoin-mining-difficulty-everything-you-need-to-know>.

²³ See e.g., “What is Ethereum,” Ethereum.org, available at <https://ethereum.org/en/what-is-ethereum/>.

applications (“dApps”). Much of this is done via tokens that represent a diverse range of digital assets, including vouchers, I Owe Yous (“IOUs”), or even tangible objects.²⁴ The term “smart contract” is used to describe a computer program or a transaction protocol that is intended to automatically execute actions according to the terms of a contract or an agreement.²⁵ A smart contract does not require human intervention to work (beyond its deployment and the triggering of specific actions). A key objective of smart contracts is to reduce reliance on trusted third-party intermediaries. A smart contract is an executable piece of code running on the blockchain that facilitates, executes, and enforces agreements between untrusted parties without the involvement of a centralized authority.²⁶

22. On the Ethereum blockchain, smart contracts enable automating transactions and agreements without requiring intermediaries. Once the conditions specified in the contract are met, the contract is triggered and automatically enforces and executes the terms.²⁷ These contracts are designed to be immutable and transparent, meaning that (i) once deployed, they typically cannot be changed, and (ii) anyone can verify their execution and review the source code underlying the contract.²⁸

23. Ethereum’s smart contract functionality and Proof of Stake (“PoS”) system allow for a wide array of use cases, including decentralized finance (“DeFi”), voting and governance systems, non-fungible tokens (“NFTs”), and more. DeFi, in particular, has seen marked growth since its introduction on Ethereum.

- a. DeFi applications aim to recreate financial services often found in traditional finance (e.g., borrowing, lending, trading) but in a decentralized and permissionless manner. Using smart contracts, users can engage in peer-to-peer transactions without relying on banks or other financial intermediaries.

²⁴ See e.g., “Introduction to smart contracts,” Ethereum.org, available at <https://ethereum.org/en/smart-contracts/>.

²⁵ Ibid.

²⁶ Ibid.

²⁷ More specifically, smart contracts are programs that are hosted and executed on a blockchain network. Smart contracts are not self-executing, meaning they require an external account to trigger their functions. Once triggered, a smart contract’s terms are executed as code running on a blockchain. See e.g., “Introduction to smart contracts,” Ethereum.org, available at <https://ethereum.org/en/smart-contracts/>; See also, Levi, Stuart D. and Lipton, Alex B, “An Introduction to Smart Contracts and Their Potential and Inherent Limitations,” Harvard Law School Forum on Corporate Governance, May 26, 2018, available at <https://corpgov.law.harvard.edu/2018/05/26/an-introduction-to-smart-contracts-and-their-potential-and-inherent-limitations/>; “What Are Smart Contracts and How Do They Work,” Chainlink, available at <https://chain.link/education/smart-contracts#frequently-asked-questions-faq-on-smart-contracts>.

²⁸ See e.g., “Introduction to smart contracts,” Ethereum.org, available at <https://ethereum.org/en/smart-contracts/>.

- b. Voting and governance systems facilitate the creation of Decentralized Autonomous Organizations (“DAOs”), which are member-governed communities or protocols where token holders can propose and vote on changes.²⁹ Governance tokens in projects like Uniswap let holders vote using UNI tokens.³⁰ Holders vote on issues such as fee changes, liquidity incentives, and protocol upgrades.
- c. NFTs have diverse applications, including arts and collectibles, music and digital ownership, in-game items, and metaverse assets.³¹

24. Since the launch of the Ethereum platform in 2015, many different token contracts have been developed that comply with the technical standards of the Ethereum network. While estimates vary, as of January 2025, the website Etherscan identifies 1,461,622 token contracts that have been launched on the Ethereum blockchain.³²

25. Ethereum’s ability to host and run smart contracts is powered by the Ethereum Virtual Machine (EVM),³³ which executes the contract code, while its PoS mechanism ensures network security and consensus for processing these transactions.³⁴ In PoS systems, any entity that meets specific requirements can operate a “validator,” enabling them to propose and verify blocks in proportion to the amount of cryptocurrency they have staked on the network. On Ethereum, a validator node must stake a minimum of 32 ETH and run specific “client” software to operate.³⁵

26. Validators are essential to the Ethereum protocol’s operation, as they maintain its operation and security by building and confirming new blocks. At a high level, new blocks on Ethereum are built via the following process:

²⁹ See e.g., “What is a DAO,” Ethereum.org, available at <https://ethereum.org/en/dao/>.

³⁰ See e.g., “Beginners’ Guide to Voting,” Uniswap Docs, available at <https://docs.uniswap.org/concepts/governance/guide-to-voting>.

³¹ See e.g., “What is an NFT? Non-fungible tokens explained,” Kraken.com, available at <https://www.kraken.com/learn/what-are-non-fungible-tokens-nft>; See also, Mitchell Clark, “NFTs, explained,” The Verge, June 6, 2022, available at <https://www.theverge.com/22310188/nft-explainer-what-is-blockchain-crypto-art-faq>.

³² See “Token Tracker (ERC-20),” Etherscan, available at <https://etherscan.io/tokens>. Accessed on January 17, 2025.

³³ The Ethereum Virtual Machine is a decentralized virtual environment that executes code consistently and securely across all Ethereum nodes. See “Ethereum Virtual Machine (EVM),” Ethereum.org, available at <https://ethereum.org/en/developers/docs/evm/>.

³⁴ Ethereum transitioned to a Proof of Stake (PoS) mechanism through an upgrade known as “The Merge,” which was completed on September 15, 2022. See, “The Merge,” Ethereum.org, July 24, 2024, available at <https://ethereum.org/en/roadmap/merge/>.

³⁵ See e.g., “What is Staking?” Ethereum.org, available at <https://ethereum.org/en/staking/>.

- a. A block proposer is selected and is responsible for gathering transactions, assembling them into a block, and proposing the block to the network.³⁶ The transactions included are selected from the mempool and must meet certain validity requirements, such as valid and correct signatures, sufficient tokens for transfer, and adequate “gas” fees. A gas fee is the cost paid by users to compensate validators for the computational work required to process transactions and execute smart contracts on the network.³⁷ Gas fees are determined by network demand, increasing during periods of high activity as users compete to have their transactions prioritized.³⁸
- b. Other validators on the network must then attest to the block by voting on its validity. New blocks require more than two-thirds of validators to verify their validity before they can be added to the blockchain.³⁹
- c. Lastly, the block is finalized and the transactions within the block are considered permanent and unalterable.

27. New blocks are added on average every 12 to 14 seconds.⁴⁰ This is considerably faster than Bitcoin’s block time (10 minutes), but it is still far from instantaneous. The latency between a user initiating a transaction and its confirmation has inspired the development of “Layer 2” solutions. These additional protocols are built on top of Ethereum’s base layer (“Layer 1”)⁴¹ and provide a partial fix to the scalability and latency issues of Ethereum. Layer 2 solutions can enhance scalability, reduce latency, and lower transaction fees by processing transactions off the base layer. For instance, “Optimistic Rollups,” such as Arbitrum and Optimism, are Layer 2 solutions that aggregate multiple transactions into a single “rollup” and periodically submit

³⁶ See e.g., “Blocks,” Ethereum.org, available at <https://ethereum.org/en/developers/docs/blocks/>; The block proposer is determined proportionally to their staked ETH in the network, which can be either self-owned or delegated to a validator. This distinguishes a PoS blockchain from Bitcoin’s PoW approach, where block builders are determined through computational power. The key advantage of PoS is its significantly lower energy consumption compared to PoW, which relies on computers solving complex mathematical and cryptographic problems that require substantial energy.

³⁷ See e.g., “GAS AND FEES,” Ethereum.org, available at <https://ethereum.org/en/developers/docs/gas/>.

³⁸ Ibid.

³⁹ See e.g., “HOW A TRANSACTION GETS EXECUTED IN ETHEREUM POS,” Ethereum.org, available at <https://ethereum.org/en/developers/docs/consensus-mechanisms/pos/>.

⁴⁰ See e.g., “BLOCK TIME,” Ethereum.org, available at <https://ethereum.org/en/developers/docs/blocks/#block-time>.

⁴¹ The base layer, or “Layer 1” refers to the main Ethereum blockchain.

compressed data to Ethereum.⁴² In essence, they function as separate networks that submit transaction data to the original blockchain (in this case, Ethereum) for validation and security. Although these additional protocols can improve certain aspects, they also have potential drawbacks. For example, developers and users face a more fragmented environment where different applications or asset types might run on separate chains. These complexities include bridging between multiple chains, dealing with separate chain IDs, splitting liquidity across different environments, or introducing certain operational risks.

- a. *Bridging between Chains.* When interacting with these various layers, a user might have to move tokens and assets between them through “bridges.” A bridge is essentially a system that locks tokens on one chain and issues corresponding tokens on another, allowing for asset transfers between layers or protocols.⁴³ However, bridges require smart contracts, and have been vulnerable to hacks and theft.⁴⁴
- b. *Separate Chain IDs.* Each blockchain typically has its own unique chain identifier (chain ID).⁴⁵ Each may also have its own transaction formats, node software, programming environments, and token standards. As a result, developers have to write and maintain code in different contexts. Contracts or programs that run on one chain may not be directly compatible with another. Building an application that can seamlessly interact across multiple chains or layers becomes more difficult and error-prone because developers have to account for different protocols and standards.
- c. *Liquidity Fragmentation.* Aggregate market liquidity becomes fragmented when assets and trading activity are spread across multiple networks. For instance, if a popular token is traded on Ethereum’s main chain and also on multiple Layer-2 networks, the end result is separate liquidity pools. Each environment might have

⁴² See e.g., “OPTIMISTIC ROLLUPS,” Ethereum.org, available at <https://ethereum.org/en/developers/docs/scaling/optimistic-rollups/>.

⁴³ See e.g., Robert Stevens, “What Are Blockchain Bridges and How Do They Work?,” Coindesk, May 11, 2023, available at <https://www.coindesk.com/learn/what-are-blockchain-bridges-and-how-do-they-work>.

⁴⁴ For example, in 2022, the Wormhole bridge was hacked via an exploited vulnerability in the smart contract code and \$325 million was lost. See “The risks of using cross-chain bridges,” Cointelegraph, April 2024, available at <https://cointelegraph.com/learn/articles/risks-of-using-cross-chain-bridges>.

⁴⁵ See e.g., “What is the Chain ID in Blockchain?,” Nadcab Labs, available at <https://www.nadcab.com/blog/chain-id-in-blockchain>.

its token variants (referred to as “wrapped” versions) and different prices.

Traders and liquidity providers must choose where to deploy their capital.⁴⁶ This dispersal means shallower liquidity in each place, potentially leading to more volatile prices and higher price slippage for traders. It also creates inefficiencies for users, who must decide which token version to use and where to move their assets to find the most optimal trading conditions.

- d. *Operational Risks in Layer 2 Solutions.* Layer 2 solutions typically have a single team operating the “sequencer,” which is a node that organizes transactions on a Layer 2 protocol.⁴⁷ A sequencer is responsible for ensuring that transactions are in the correct order and are valid.⁴⁸ Sequencers also often rely on trust – for instance – on Base and ZKsync Era, code upgrades are implemented and distributed without delay.⁴⁹ As a result, if a malicious upgrade is approved, in principle, there is a risk for users to lose their funds. Furthermore, all Layer 2 solutions powered by optimistic rollups have a “challenge window” of up to seven days, during which transactions are not yet finalized on the base layer and can be challenged.

B. The Solana Decentralized Network

28. Solana is a blockchain designed for speed and efficiency.⁵⁰ Unlike Bitcoin’s relatively long block time and Ethereum’s use of Layer 2 solutions, Solana is designed to provide high

⁴⁶ An Automated Market Maker (AMM) an algorithmic agent relying on crowdsourced pools of cryptocurrencies or tokens locked in a self-executing smart contract in order to supply liquidity and mutualize risk (and return) in an electronic market. The smart contract paradigm allows transactions to be conducted peer-to-contract without central controlling or clearing mechanisms. The AMM smart contracts pool liquidity where those who want to be a market maker (i.e., liquidity providers) get liquidity tokens representing their stake in the pool. *See e.g.*, “What Are Automated Market Makers (AMM)?,” Cryptopedia, October 2023, available at <https://www.gemini.com/cryptopedia/amm-what-are-automated-market-makers>

⁴⁷ *See e.g.*, “What Are Sequencers in Layer 2 Protocols Such as Optimism, Arbitrum, and Base?” Unchained Crypto, available at <https://unchainedcrypto.com/what-are-sequencers-in-layer-2-protocols-such-as-optimism-arbitrum-and-base/>.

⁴⁸ *Ibid.*

⁴⁹ “ZKsync Era,” L2BEAT, available at <https://l2beat.com/scaling/projects/zksync-era> and <https://l2beat.com/scaling/projects/base>.

⁵⁰ Solana was first conceptualized as a high-performance blockchain in 2017 by Anatoly Yakovenko in a whitepaper that aimed to address scalability issues inherent in existing networks such as Bitcoin and Ethereum. Mr. Yakovenko was subsequently joined by Greg Fitzgerald and Stephen Akridge, his former Qualcomm colleagues, and started developing a prototype of a new blockchain based on the whitepaper. The founding team was joined by Raj Gokal in 2018, and in that same year, Solana Labs was established as a for-profit entity responsible for developing the blockchain and its underlying technology. In March 2020, Solana launched its Mainnet Beta, allowing users and

throughput and low latency directly at the base layer. It achieves this by integrating its software with the capabilities of modern hardware, using a single, unified network where all applications run on the same chain.⁵¹ This avoids complexities primarily associated with Ethereum and the Layer 2 solutions discussed above.

29. While Layer 2 solutions exist on Solana, they are not necessary for increasing throughput.⁵² Users can conduct transactions on the main network without requiring cross-chain interactions to manage fee differences or seek faster transaction confirmation times. Every application deployed on Solana can interact directly with every other application running on the same ledger. Users hold their tokens in accounts directly on this single blockchain. By requiring the network to agree on a single state, liquidity can be consolidated in one place, eliminating the need for bridges and separate chain IDs.

30. The unified architecture of the Solana blockchain offers several important benefits:

- a. Instead of navigating complex bridges, learning multiple environments, and splitting assets across various platforms, users can do everything directly on the Solana blockchain. This design choice streamlines interactions and aims to improve security and enhance overall efficiency.
- b. Solana is notable for its fast block production (low block time)—around 400 milliseconds per block—and the ability to handle thousands of transactions per second. Transaction fees are low, often just fractions of a penny.⁵³ In contrast,

developers to interact with the network. The first block, also called the “genesis block,” was created on March 16, 2020, which marked the official launch of the live environment. *See*, “Solana: A new architecture for a high performance blockchain,” available at <https://solana.com/solana-whitepaper.pdf>; “June Newsletter,” Solana Foundation, July 5, 2020, available at <https://solana.com/news/june-newsletter>; The Solana Foundation was established in June 2020 as a Switzerland-based non-profit organization with a goal of supporting growth and decentralization of the Solana blockchain. *See*, “Mission Statement,” Solana Foundation, available at <https://solana.org/about>.

⁵¹ When we refer to Solana, we are referring to the “mainnet,” which is the active network, as opposed to the “testnet,” which refers to the public network used to test smart contracts and apps before they are deployed to the mainnet.

⁵² *See* “Layer 2 & Scaling Solutions,” Solana Compass, available at <https://solanacompass.com/projects/category/infrastructure/scaling>; *See also*, “Layer 2 On Solana? Possible But Irrelevant, Says Co-founder – Here’s Why,” Samuel Edyme, available at <https://bitcoinist.com/layer-2-solana-possible-irrelevant-says-co-founder/>.

⁵³ The Solana website reports the median fee per transaction as \$0.00064. *See*, <https://solana.com/>, last accessed January 17, 2025.

transaction fees on Ethereum and Bitcoin averaged \$5.31 and \$3.09, respectively, in December 2024.⁵⁴

- c. On Solana, most apps run in the same global state, enabling them to interact smoothly with one another, eliminating the need for developers to build infrastructure for bridging tokens between multiple layers. Users can move seamlessly between various applications because all state and liquidity live under one unified protocol.

1. How transactions work

31. A typical Solana transaction begins with a user signing a message using their private key. Public keys serve as unique account identifiers, much like addresses in Bitcoin or Ethereum. Private keys prove ownership and authorization over these accounts. To send a transaction, the user's wallet assembles all the necessary parameters—such as which accounts to read or write, which programs to call, and a recent blockhash⁵⁵ to prevent replay attacks.^{56,57}

32. The transaction is sent to a designated validator called the “leader.” On Solana, leaders take turns producing blocks according to a pre-determined schedule that changes every epoch (a Solana epoch is approximately two days long). Once the leader receives the transaction, it is executed and included in the next block if it is valid and does not conflict with other

⁵⁴ See e.g., “Average Transaction Fee Chart,” Etherscan.io, available at <https://etherscan.io/chart/avg-txfee-usd>; “Bitcoin Average Transaction Fees,” BitInfoCharts, available at <https://bitinfocharts.com/comparison/bitcoin-transactionfees.html>.

⁵⁵ A “blockhash” refers to the last Proof of History (PoH) hash for a “slot.” Since Solana uses PoH as a trusted clock, a transaction's recent blockhash can be thought of as a timestamp. Slots refer to the period of time for which each leader ingests transactions and produces a block. Collectively, slots create a logical clock. Slots are ordered sequentially and non-overlapping, comprising roughly equal real-world time as per PoH. See, “Transaction Confirmation & Expiration,” Solana.com, available at <https://solana.com/docs/advanced/confirmation>.

⁵⁶ Replay attacks refer to malicious acts where an “Attacker is able to replay previously captured messages (between a legitimate Claimant and a Verifier) to masquerade as that Claimant to the Verifier or vice versa.” See e.g., “Computer Security Resource Center Glossary,” National Institute of Standards and Technology, U.S. Department of Commerce, available at https://csrc.nist.gov/glossary/term/replay_attack.

⁵⁷ An acceptable blockhash must be sufficiently recent, but it does not necessarily have to be the hash of the very last block created. The inclusion of a recent blockhash helps to ensure that the transaction is submitted and confirmed within a relatively short timeframe. If a blockhash is deemed to be too old, the transaction becomes invalid, which typically happens after a couple of minutes as new blocks are added to the chain. See e.g., “Transaction Confirmation & Expiration,” Solana.com, available at <https://solana.com/docs/advanced/confirmation>.

transactions. After execution, the leader broadcasts the block to the network's other validators, who verify and confirm the results.⁵⁸

33. On Bitcoin and Ethereum, pending transactions float in a public mempool awaiting inclusion by miners or validators. Solana forgoes the traditional mempool model. Instead, transactions are sent directly to the leader for each slot (a brief period when that leader can produce a block). This approach reduces latency and ensures the network can quickly adapt to changing load conditions.

34. Solana's ledger architecture is engineered for speed. It uses the QUIC protocol⁵⁹—originally developed by Google—to handle large volumes of incoming transactions efficiently. Solana's "Gulf Stream" forwarding mechanism sends transactions straight to the expected future leaders.⁶⁰ This combination of optimizations enables the network to maintain performance and efficiency even during periods of heavy load.⁶¹

35. A key difference from Ethereum's model lies in how Solana manages and updates the state of its ledger at any given time. Solana transactions must explicitly declare which accounts they will read and write.⁶² This allows the validator to identify non-overlapping sets of accounts and process multiple transactions in parallel without conflicts. Parallelization is essential for

⁵⁸ See e.g., Jeff Albus, "The lifecycle of a Solana transaction," Blockworks, July 23, 2024, available at <https://blockworks.co/news/light-speed-newsletter-solana-transaction-lifecycle>.

⁵⁹ QUIC (Quick UDP Internet Connections) is a transport layer network protocol initially developed by Google and later standardized by the Internet Engineering Task Force (IETF). It uses the User Datagram Protocol (UDP) and is designed to provide faster, more reliable, and more secure connections compared to traditional protocols like Transmission Control Protocol (TCP).

⁶⁰ On Solana, after a transaction is signed and before it is confirmed in a new block, it is forwarded to a Remote Procedure Call (RPC) provider. Essentially, RPC providers act as intermediaries between applications (e.g., DeFi) and the validators that build new blocks. RPCs are similar to validator nodes, in that RPCs run the same software as full validators. This allows RPCs to "accurately simulate transactions and maintain an up-to-date view of the current state" of the Solana blockchain. RPCs differ from full validator nodes in that they do not hold a stake in the network. In other words, RPCs do not have to stake SOL tokens in order to run the validator software and as such, RPCs cannot vote or build blocks. Block building is conducted only by certain nodes, referred to as "leaders." See e.g., "Solana Executive Overview," available at <https://www.helius.dev/blog/solana-executive-overview>; "Gulf Stream: Solana's Mempool-less Transaction Forwarding Protocol," Solana Foundation, June 19, 2019, available at <https://solana.com/news/gulf-stream--solana-s-mempool-less-transaction-forwarding-protocol>.

⁶¹ Solana's protocol requires that a single validator be designated as the "leader" for every block created. Once an RPC receives a transaction to be included in a block, it sends that transaction to the leader, who organizes transactions into blocks. Validators with a greater SOL stake have a higher chance of becoming a leader. Importantly, the schedule of leaders is known ahead of time. The clear delineation in roles between RPCs and leaders "allows validators to execute transactions ahead of time, reduce confirmation times, switch leaders faster, and reduce the memory pressure on validators from the unconfirmed transaction pool." See, "Gulf Stream: Solana's Mempool-less Transaction Forwarding Protocol," Solana Foundation, June 19, 2019, available at <https://solana.com/news/gulf-stream--solana-s-mempool-less-transaction-forwarding-protocol>.

⁶² See e.g., "Transaction and Instructions," Solana.com, available at <https://solana.com/docs/core/transactions>.

high throughput and helps Solana scale with modern multicore processors. Solana transactions also feature pre-declared fees and a “compute” budget⁶³ (maximum computational resources a transaction is permitted to consume during its execution), which enables more efficient scheduling of transactions by providing advance notice of resource requirements.

2. Proof of History (PoH) and Consensus

36. One of Solana’s distinctive features is Proof of History, a cryptographic clock that provides a verifiable ordering of events. Instead of relying solely on validator communication to agree on time, Solana validators maintain a cryptographic sequence of hashes that proves the time elapsed between events. That is, each transaction has a unique identifier that depends on the previous transaction’s identifier. This time series dependency allows one to uniquely identify the timing of transactions, which reduces the complexity of coordination, allowing the network to settle on an order of transactions more quickly and with fewer communication rounds.⁶⁴

37. This hybrid approach enables Solana to achieve high transaction throughput and comparatively low latency, which allows for the processing of thousands of transactions per second. Although Solana’s maximum recorded transactions per second (TPS) is 7,229, it can theoretically handle up to 65,000 TPS, with a potential to scale up to 710,000 TPS.⁶⁵ During the first half of 2024, Solana averaged approximately 2,200 TPS⁶⁶ and on January 17, 2025, it averaged over 4,000.⁶⁷ For comparison, Bitcoin’s throughput is 7 TPS and Ethereum averages approximately 12 TPS. As for processing speed, an approximate time for a transaction to reach “finality,” i.e., to be confirmed on the blockchain, is 0.4 seconds for Solana compared to

⁶³ See e.g., “Compute Budget,” Solana.com, available at <https://solana.com/docs/core/fees>.

⁶⁴ This time series dependency also serves to track the leader schedule, ensuring that each leader produces blocks in the correct order. Specifically, the network is aware of which leader is scheduled to produce a block, and the PoH mechanism prevents a malicious leader from submitting a pre-packaged block, thereby allowing the honest leader to produce valid transactions and the network to accept them in a timely manner. Additionally, PoH enables the next leader to synchronize its block production with the network’s acceptance of its blocks, reducing the latency between leader transitions.

⁶⁵ For comparison, the payment network Visa has the capacity to execute more than 65,000 transactions per second. See Mustafa Bedawala and Arjuna Wijeyekoon, “A deep dive on Solana, a high-performance blockchain network,” Visa, available at <https://usa.visa.com/solutions/crypto/deep-dive-on-solana.html>.

⁶⁶ See “Solana’s Average TPS Reaches 2200 With Over 80% Transaction Success Rate in H1 2024,” Binance News, available at <https://www.binance.com/en/square/post/2024-06-28-solana-s-average-tps-reaches-2200-with-over-80-transaction-success-rate-in-h1-2024-10085491122530>.

⁶⁷ See Solscan.io, last accessed January 17, 2025.

approximately 3-5 seconds for Algorand, 15 minutes for Ethereum, and up to 60 minutes for Bitcoin.⁶⁸

38. Solana employs Tower BFT for consensus, which is a variant of Byzantine Fault Tolerance.⁶⁹ As Anatoly Yakovenko describes, “Tower BFT leverages Solana’s PoH as a clock before consensus to reduce messaging overhead and latency.”⁷⁰ Under this protocol, validators “vote” on blocks they consider valid. This approach “allows the network to continuously stream blocks without stalling the ledger until a supermajority observes the same ledger.”⁷¹ As time passes, the network finalizes the chain with increasing certainty, making confirmed transactions highly unlikely to be reversed.

3. Economic model

39. Like Ethereum, the Solana network relies on transaction fees to deter network spam and reward block producers. The presence of transaction fees discourages excessive or malicious activity on the network. Transaction fees also compensate the node operators who verify and record transactions. In addition to the transaction fees, the network requires a “storage fee” referred to as “rent,” which is “the fee deposited into every Solana account to keep its associated data available on-chain.”⁷² This rent is reclaimable when an account is closed but incentivizes a more efficient use of the network by removing inactive accounts that are not maintaining a minimum balance.

40. Transaction fees are distributed as rewards to the validators who create new blocks and secure the network. Validators on Solana stake SOL tokens, which means they commit a certain amount of SOL as a security deposit, run validator software, and have “voting rights” proportional to the amount staked.⁷³ This staking process aligns the validators’ financial interests

⁶⁸ See Mustafa Bedawala and Arjuna Wijeyekoon, “A deep dive on Solana, a high performance blockchain network,” Visa, available at <https://usa.visa.com/solutions/crypto/deep-dive-on-solana.html>; See also “Single Slot Finality,” Ethereum.org, May 2024, available at <https://ethereum.org/en/roadmap/single-slot-finality/>.

⁶⁹ See e.g., Yakovenko, Anatoly, “Tower BFT: Solana’s High Performance Implementation of PBFT,” Medium, July 2019, available at <https://medium.com/solana-labs/tower-bft-solanas-high-performance-implementation-of-pbft-464725911e79>

⁷⁰ Ibid.

⁷¹ Ibid.

⁷² See “Fees on Solana,” Solana.com, available at <https://solana.com/docs/core/fees>.

⁷³ See e.g., “Staking Overview,” Solana.com, available at <https://solana.com/staking>.

with the network's health and deters malicious behavior, as attempts to compromise the network could result in the loss of their staked funds.

41. Users who do not run their own validator node can opt to delegate SOL tokens to one who does. This process is referred to as delegated staking.⁷⁴ Users who opt to delegate their stake can select validators based on factors such as a validator's commission rate, uptime and reliability, and overall reputation.⁷⁵ Upon selecting a validator and confirming delegation of the tokens, the chosen validator is assigned the voting rights of the delegated SOL tokens, but does not take custody of the tokens themselves. From the user's perspective, there is an activation period that spans approximately one epoch, during which time, the stake activates and starts earning rewards. Similarly, if a user decides to un-delegate, the tokens undergo a deactivation period that lasts one epoch, before they become available for withdrawal.⁷⁶

42. In addition, validators benefit from newly issued tokens granted as staking rewards. This system not only encourages participation but also supports ongoing investment in the network's stability. Ultimately, staking helps secure the network against potential attacks while ensuring its continued growth, functionality, and economic resilience. As of January 2025, there were over 1,400 validators on Solana.⁷⁷

4. Decentralization

43. Decentralization is a founding principle of blockchain networks. By design, most blockchain networks operate via a connected network of nodes, all of which independently verify transactions and record the network's history without the need for a centralized entity.

44. Moreover, becoming a validator on Solana is a completely permissionless process, allowing any entity to participate in verifying and securing the network. This is a significant departure from many other Layer 1 and Layer 2 blockchain networks, which often employ more restrictive selection processes. The method by which validators are chosen impacts the network's governance, as it ultimately determines who holds control. In contrast, Solana's

⁷⁴ Ibid.

⁷⁵ See e.g., "Validators," Solana Beach, available at <https://solanabeach.io/validators>.

⁷⁶ See "Staking and Inflation FAQ," Solana Foundation, available at <https://solana.com/staking>.

⁷⁷ "Validators," Solscan.io, available at <https://solscan.io/validator> last accessed January 17, 2025.

permissionless approach ensures decentralization, as each participating entity collectively contributes to the network's operation and decision-making.

45. Decentralization is more than just a feature of blockchains; it is fundamental to maintaining the integrity and security of the network. Decentralized systems aim to address the potential vulnerabilities inherent to centralized systems, including the existence of single points of failure, susceptibility to hostile attacks, and malicious behavior by the single entity or small group controlling the system. Decentralization in blockchains addresses these concerns by dispersing control across a diverse set of network participants. Additionally, it is essential for the trustless and permissionless nature of blockchain systems and enables robust consensus mechanisms. Therefore, academics and practitioners usually take a strong interest in the decentralization measures of blockchains and how they impact network performance.⁷⁸

46. A common measure used to quantify the degree of decentralization is the Nakamoto Coefficient, which was first proposed by Balaji Srinivasan and Leland Lee.⁷⁹ The larger the coefficient, the more decentralized the blockchain. The Nakamoto Coefficient describes the minimum number of compromised validators (nodes) required to disrupt a blockchain by denying the consensus mechanism required for new block production.⁸⁰ In other words, it is the smallest number of independent entities that can act collectively to shut down a blockchain. In Solana's PoS based network, this coefficient measures the fewest number of nodes that control at least 33.4% of the voting power.⁸¹ As of January 2025, Solana's Nakamoto Coefficient is 18, which ranks above many competing PoS blockchains.⁸²

47. Another insight into the decentralization of a blockchain network is the geographic diversity of its validators. Solana has a globally distributed network of validators and data centers, which helps mitigate the negative impact of any local disruptions, outages and

⁷⁸ See e.g., Gencer, Adem Efe et al, "Decentralization in Bitcoin and Ethereum Networks," March 29, 2018, available at <https://arxiv.org/abs/1801.03998>.

⁷⁹ Balaji Srinivasan and Leland Lee, "Quantifying Decentralization," Medium, July 28, 2017 ("Srinivasan and Lee (2017)"), available at <https://news.earn.com/quantifying-decentralization-e39db233c28e>.

⁸⁰ Srinivasan and Lee (2017), ("In this post we propose the minimum Nakamoto coefficient as a simple, quantitative measure of a system's decentralization, motivated by the well-known Gini coefficient and Lorenz curve. The basic idea is to (a) enumerate the essential subsystems of a decentralized system, (b) determine how many entities one would need to be compromised to control each subsystem, and (c) then use the minimum of these as a measure of the effective decentralization of the system. The higher the value of this minimum Nakamoto coefficient, the more decentralized the system is.").

⁸¹ The Nakamoto Coefficient is the smallest k such that $\sum_{i=1}^k S_i > 1/3$, where S_i is the stake of the i -th validator sorted by descending order.

⁸² See e.g., "Nakamoto Coefficients," available at <https://nakaflow.io/>.

regulatory interventions. The network has around 4,500 nodes in total and over 1,400 validator nodes.⁸³ As of November 2024, Solana validator nodes spanned 37 countries.⁸⁴

48. Lastly, the distribution of staked assets among validators can provide yet another measure of decentralization. In practice, some validators are more likely to have larger stakes because they provide services to users. However, as of January 17, 2025, no validator holds more than 3.4% of SOL’s total stake.⁸⁵ The list of validators, their corresponding stake, and the leader schedule are all publicly available via Solana blockchain explorers.⁸⁶

III. THE SOL TOKEN AND MARKET METRICS

49. We now turn to analyzing the market for the SOL token. In subsection A, we begin with a review of the relevant “tokenomics” of SOL. Broadly speaking, tokenomics refers to the design of the economic systems that govern cryptocurrencies.⁸⁷ Key aspects of tokenomics include the specifics of initial token distribution and subsequent new token issuance.

Understanding the systems that govern SOL provides important background, as it describes both the supply of SOL tokens and supply changes over time.

50. In subsection B, we then provide a detailed review of the SOL market. We first describe the venues on which SOL is traded, including both centralized exchanges (“CEXs”) and decentralized exchanges (“DEXs”). The structure of each venue is different and both are important for understanding price formation.

51. In subsection C, we first provide a brief discussion of regulated SOL products that exist in Europe and Brazil. These products provide investors with regulated access to SOL without the need to directly purchase, manage, and custody cryptocurrencies. We then analyze the market structure of SOL and provide insight into the market characteristics potentially relevant to future U.S. regulated product approval. These include key measures of market quality, such as order book liquidity, effective spreads, and trade costs, as well as the degree to which SOL

⁸³ See “Measuring Solana’s Decentralization: Facts and Figures,” Helius, Nov. 18, 2024, available at <https://www.helius.dev/blog/solana-decentralization-facts-and-figures>.

⁸⁴ See e.g., “Solana Validator Metrics,” Block Logic, available at <https://www.validators.app/>.

⁸⁵ See e.g., “Validators,” Solana Beach, available at <https://solanabeach.io/validators>.

⁸⁶ See e.g., “Solana Explorer,” available at <https://explorer.solana.com/>.

⁸⁷ See e.g., Ralf Wandmacher, “Cryptofinance and Mechanisms of Exchange,” available at https://link.springer.com/chapter/10.1007/978-3-030-30738-7_7.

prices are correlated across markets. We find that the SOL market is large, liquid, and with characteristics that are by several measures broadly comparable to the market for BTC and ETH.

A. Tokenomics

52. Developers' choice of tokenomic attributes can depend on the role that a specific crypto asset plays in the broader blockchain ecosystem. Here, a fundamental distinction lies between tokens that exist natively on blockchains ("native cryptocurrencies," "intrinsic tokens," "protocol tokens," or simply "coins") and tokens built on existing blockchains.⁸⁸

- a. Native cryptocurrencies, such as Bitcoin (BTC) on the Bitcoin blockchain, Ether (ETH) on the Ethereum blockchain, and SOL on the Solana blockchain enable and support core operations of the respective networks.⁸⁹ They are used to incentivize either miners, as in the case of Bitcoin, or validators, as in the cases of Ethereum and Solana (see, e.g., discussion above). They are also primary mediums of exchange on the respective networks.⁹⁰
- b. In contrast, "tokens" are digital assets that utilize the infrastructure of already existing blockchains. They leverage the security and the consensus mechanism of the network they rely upon in order to facilitate the proper functioning of decentralized applications on those blockchains.⁹¹ For instance, UNI on Ethereum serves as a governance token for the Uniswap decentralized exchange.⁹² As another example, LINK is used to pay for oracle services⁹³ within the Chainlink

⁸⁸ See e.g., Reyes, Carla, "The Language Landmines of Blockchain Technology and Cryptocurrency," The Cambridge Handbook on Law and Policy for NFTS, November 23, 2023 ("Reyes 2023").

⁸⁹ Reyes 2023, p. 9; See also, "Digital Assets: Cryptocurrencies vs. Crypto Tokens," Cryptopedia, available at <https://www.gemini.com/cryptopedia/cryptocurrencies-vs-tokens-difference>; "Defining Ether and Ethereum," CME Group, available at <https://www.cmegroup.com/education/courses/introduction-to-cryptocurrency-futures/ether/defining-ether-and-ethereum.html>

⁹⁰ See e.g., Reyes 2023, pp. 8-9; See also, "What is the difference between a coin and a token," Coinbase, available at <https://www.coinbase.com/learn/crypto-basics/what-is-the-difference-between-a-coin-and-a-token>

⁹¹ See e.g., Reyes 2023, p. 9; See also, "PYMNTS Crypto Basics Series: What's a Native Token, Non-Native Token and White-Label Crypto?" PYMNTS, available at <https://www.pymnts.com/cryptocurrency/2022/pymnts-crypto-basics-series-whats-a-native-token-non-native-token-and-white-label-crypto/>; "Crypto Tokens: Utility, Governance and Security Tokens Explained," Ledger Academy, available at <https://www.ledger.com/academy/fungible-token-categories-explained>.

⁹² See e.g., "What is Uniswap? A Breakdown," Cryptopedia, available at <https://www.gemini.com/cryptopedia/uniswap-decentralized-exchange-crypto-defi>.

⁹³ Oracle services provide external data (i.e., data that exists "off-chain," for example, price data related to other crypto or traditional financial assets) to smart contracts that operate on the blockchain. See e.g., "Blockchain

oracle network that links smart contracts to real-world data.⁹⁴ Some of the most frequently traded or exchanged tokens are stablecoins that reflect the value of a US Dollar on a blockchain.

53. The method of initial distribution of tokens varies and depends on the objectives that each blockchain attempts to achieve. The primary goal of Bitcoin was decentralization, and the newly issued tokens were only obtainable through active participation in mining.⁹⁵ In contrast, Ethereum was a resource-intensive project, and initial ETH tokens were sold through a presale to finance research and development of the blockchain.⁹⁶

54. Solana's initial token distribution totaled 500 million SOL and was conducted in several stages. Four private sales occurred between 2018 and 2020, from which 180.15 million tokens were sold at prices ranging from \$0.04 to \$0.25 per token.⁹⁷ In March 2020, Solana then conducted a public token auction, where 8 million SOL tokens were sold at a clearing price of \$0.22 per token. The remainder of SOL tokens were distributed to the grant pool,⁹⁸ team members, the Solana Foundation, and a Community Reserve Fund.⁹⁹ Tokens of the team members were subject to a vesting schedule, and were fully vested as of July 2023.¹⁰⁰

55. Beyond the initial distribution, the method and scope of new token issuance also varies across blockchain protocols. For example, one of Bitcoin's design principles was a provision of absolute scarcity of Bitcoin supply. New tokens are issued at a decreasing rate until the total number of Bitcoin in circulation reaches approximately 21 million. The issuance of new tokens plays a crucial role as it serves as an incentive for miners to expend resources to secure the

Oracles Explained: Decentralized Oracles in DeFi," Gemini Cryptopedia, available at <https://www.gemini.com/cryptopedia/crypto-oracle-blockchain-overview>

⁹⁴ See e.g., "What Is a Blockchain Oracle?" Chainlink, available at <https://chain.link/education/blockchain-oracles>.

⁹⁵ See e.g., Francesco Bruschi et al., "A scalable decentralized system for fair token distribution and seamless users onboarding," Blockchain: Research and Application, Volume 3, Issue 1(2022) ("Bruschi et al 2022").

⁹⁶ See e.g., Bruschi et al 2022; See also, Victoria Van Eyk, "Ethereum Launches Own 'Ether' Coin, With Millions Already Sold," Coindesk, September 11, 2021, available at <https://www.coindesk.com/markets/2014/07/23/ethereum-launches-own-ether-coin-with-millions-already-sold>.

⁹⁷ See e.g., <https://icodrops.com/solana/>.

⁹⁸ A "grant pool" refers to a collection of tokens managed by an entity (like a foundation or DAO) that are allocated as grants to projects or individuals developing on the blockchain.

⁹⁹ The Solana Foundation and the Community Reserve Fund "supports projects that generate public goods for the community and give other ecosystem participants the opportunity to learn from and build on [prior] work." See e.g., "Grants and Funding," Solana.org, available at <https://solana.org/grants-funding>; See also, Livello Intermedio, "Solana Tokenomics: Everything to Know," Crypto.com, available at <https://crypto.com/it/university/solana-tokenomics>.

¹⁰⁰ See e.g., "Token Unlocks," available at <https://solana.messari.io/token-unlocks>.

network. That said, the Bitcoin Whitepaper assumed that the network will be sufficiently mature by the time the predetermined number of coins enter the circulation for the incentive to be fully funded by transaction fees paid by blockchain's users.¹⁰¹

56. As opposed to Bitcoin's inelastic supply, Ethereum implemented a semi-elastic supply.¹⁰² New ETH tokens are created indefinitely, and the amount depends on the number of active validators on the network. The staking yields rise when the demand for staking decreases and the inflation rate (new coin growth rate) remains relatively low even in times when a high proportion of total token supply is committed to staking.¹⁰³ The flexibility in Ethereum's supply expansion enables the network to dynamically adjust the incentive structure that underlies the security of the blockchain.

57. Solana's token creation model is different than either Bitcoin's or Ethereum's. Instead of relying on a fixed supply cap as in Bitcoin, or issuance model determined by network activity as in Ethereum, Solana's token creation is currently fixed and predetermined but not subject to a hard cap. It involves three parameters – the initial inflation rate, that is the annualized growth rate related to the total number of SOL tokens outstanding (locked and unlocked), the disinflation rate (the rate at which inflation is reduced), and the long-term inflation rate.¹⁰⁴ At inception, the rate of supply expansion, or the initial annualized inflation rate, was 8%, which is decreased by the disinflation rate of 15%.¹⁰⁵ The long-term inflation rate was set to 1.5%, which upon reaching, will be the rate at which new SOL tokens will be issued indefinitely.¹⁰⁶ Given the initial inflation rate, and disinflation rate, the long-term inflation rate is projected to be reached approximately 10 years after Solana's launch in 2020. Solana's token creation model can also be adjusted by community consensus through governance proposals to address evolving network needs or market conditions.¹⁰⁷

¹⁰¹ Nakamoto (2008), p. 4 ("Once a predetermined number of coins have entered circulation, the incentive can transition entirely to transaction fees and be completely inflation free.")

¹⁰² See e.g., Daniel Gray, "Understanding Bitcoin and Ethereum Supply," November 28, 2023, available at <https://www.fidelitydigitalassets.com/research-and-insights/understanding-bitcoin-and-ethereum-supply>.

¹⁰³ Ibid.

¹⁰⁴ "Inflation Related Terminology," Solana.com, available at <https://solana.com/docs/economics/inflation/terminology>.

¹⁰⁵ See e.g., "Is Solana's Inflation Too High?" Helius, September 11, 2024, available at <https://www.helius.dev/blog/solana-issuance-inflation-schedule>.

¹⁰⁶ Ibid.

¹⁰⁷ See "Proposed Inflation Schedule," Solana.com, available at <https://solana.com/docs/economics/inflation/inflation-schedule>.

58. As of December 2024, the total supply is approximately 590 million SOL, representing an 18% increase since its initial launch.¹⁰⁸ This includes approximately 106 million SOL that are part of the “non-circulating supply,” which are either: (i) SOL tokens originating from investments or grants that follow a vesting schedule, or (ii) SOL tokens that are owned by Solana Labs and the Solana Foundation.¹⁰⁹

B. SOL market structure

59. Cryptocurrency markets are comprised of global digital platforms where cryptocurrencies such as Bitcoin (BTC), Ethereum (ETH), Solana (SOL), and numerous others are traded. The market for crypto assets shares many similarities with markets for traditional financial products. Foundationally, both provide places where traders gather to transact. Like traditional financial assets, much of crypto asset trading occurs on exchanges that organize and match transactions via a limit order book. However, there are important differences that make crypto markets unique.

60. Unlike traditional financial product markets, crypto markets operate around-the-clock and enable users to buy and sell digital assets in both spot and derivative markets. Another point of difference is in the degree to which markets are fragmented. Whereas equities, bonds, and other traditional financial products’ trading volume may be concentrated among one or two primary exchanges, the market for crypto assets spans many exchanges across the world. As noted above, the primary venues facilitating these operations are Centralized Exchanges (“CEXs”) and Decentralized Exchanges (“DEXs”). CEXs function similarly to traditional stock exchanges, providing order matching services and trade execution. In a CEX, buy or sell orders are placed in a central limit order book (“CLOB”). Professional liquidity providers may place resting limit orders into the CLOB for selling crypto to those who want to buy and buying crypto from those who want to sell. Liquidity providers seek to profit from the difference, or spread, between their bid price and offer price when transactions take place. The spread realized by liquidity providers

¹⁰⁸ See e.g., “Solana Tokenomics,” Solana Compass, available at <https://solanacompass.com/tokenomics>; “Supply Overview,” Solana Explorer, available at <https://explorer.solana.com/supply>.

¹⁰⁹ Ibid. We note that Multicoon Capital proposed changes to Solana’s inflation rate that would change it from a fixed rate to a market-based rate based on the percentage of SOL that is staked. See e.g., Jack Kubinec, “Solana to weigh possible SOL inflation cut,” Blockworks, January 16, 2025, available at <https://blockworks.co/news/solana-weighing-sol-inflation-cut>.

is how they are compensated for facilitating trading. On centralized exchanges, liquidity providers manage their risk by continually revising their bids and offers in response to pending order flow and new information about market conditions.¹¹⁰

61. Centralized exchanges serve as intermediaries that facilitate trading of digital assets, and unlike decentralized exchanges, they provide conversion services between fiat currencies and cryptocurrencies. Some of the most popular CEXs in the U.S. include Coinbase, Kraken, and Gemini, all of which offer advanced trading features and custodial services. Because they provide a user-friendly interface, no specific technical knowledge of underlying market structure is necessary for users to trade cryptocurrencies on these venues.

62. In contrast, DEXs operate solely with the use of smart contracts, and hence, without centralized control.¹¹¹ DEXs facilitate trading via an AMM, that is, a protocol that relies on algorithms to price assets and executes transactions given a set of logic and inputs.¹¹² The protocol relies on crowdsourced pools of cryptocurrencies or tokens locked in smart contracts in order to supply liquidity and mutualize risk (and return).¹¹³ The smart contract paradigm allows transactions to be conducted without central controlling or clearing mechanisms.

1. CEX Volumes

63. The SOL token was initially listed in April 2020 and as of January 2025, it is listed on nearly all major cryptocurrency exchanges, including exchanges that are available to U.S.-based investors. Figure 1, below, depicts the total daily traded volume of the SOL token. Traded volume of SOL increased substantially between its listing in April 2020 and the end of 2021, consistent with the broader trends of the crypto “bull-market.” Traded volume waned throughout 2022 and early 2023, potentially related to key events such as the bankruptcies of large entities

¹¹⁰ Liquidity providers face “adverse selection risk” from having their open orders hit by traders with superior information about pending order flow or significant market-moving news.

¹¹¹ Companies and foundations can sometimes manage DEXs, but their role is usually limited to providing technology. See “The Role of Decentralized Exchanges (DEXs) in DeFi,” SoluLab, available at <https://www.solulab.com/decentralized-exchanges-role-in-defi/>.

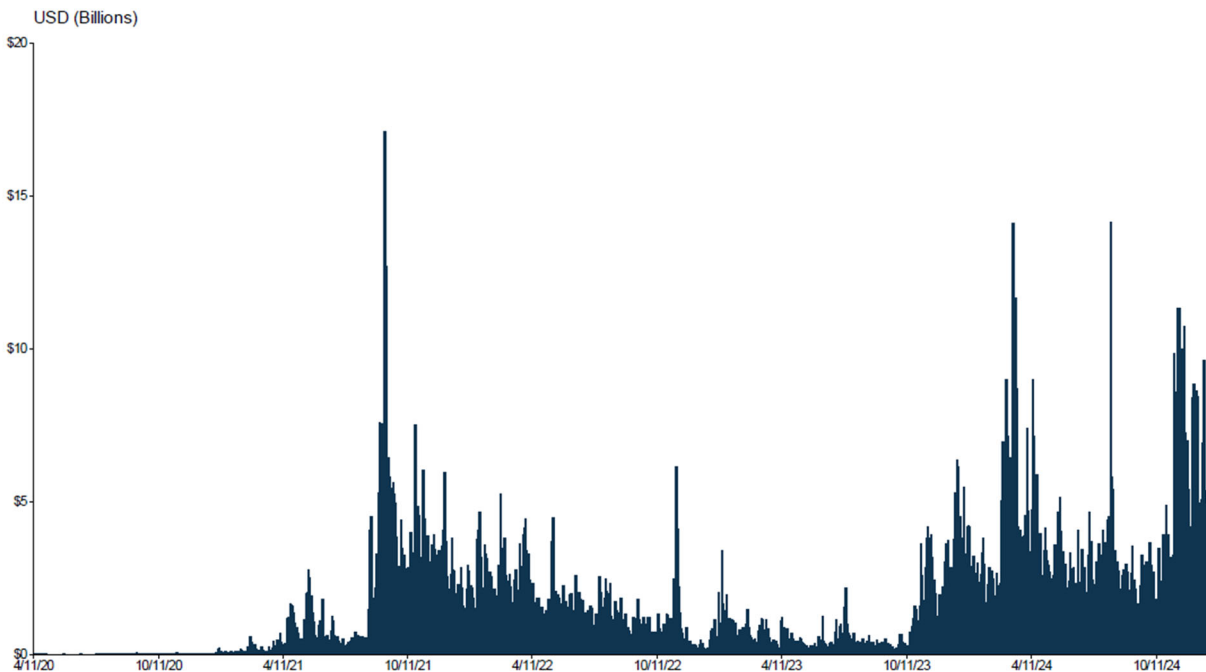
¹¹² See e.g., “What are Automated Market Makers (AMM)?,” Cryptopedia, available at <https://www.gemini.com/cryptopedia/amm-what-are-automated-market-makers>.

¹¹³ Ibid.

(e.g., FTX and Three Arrows Capital), alongside broader macroeconomic shifts.¹¹⁴ Trading volume began to recover through the latter half of 2024, and daily traded volume averaged \$5.14 billion in December 2024.

Figure 1

SOL Traded Volume, April 11, 2020 – December 31, 2024



Source: CoinMarketCap

64. The SOL market includes a substantial notional amount of perpetual futures contracts, which are derivative products that allow traders to speculate, hedge, and conduct arbitrage. Perpetual futures are unique derivative products that are especially popular in cryptocurrency trading. Unlike traditional futures contracts, perpetual futures can be held indefinitely. The two counterparties to a perpetual futures contract pay each other on an ongoing marked to market basis, where the long side pays the short side when the contract price is higher than the market

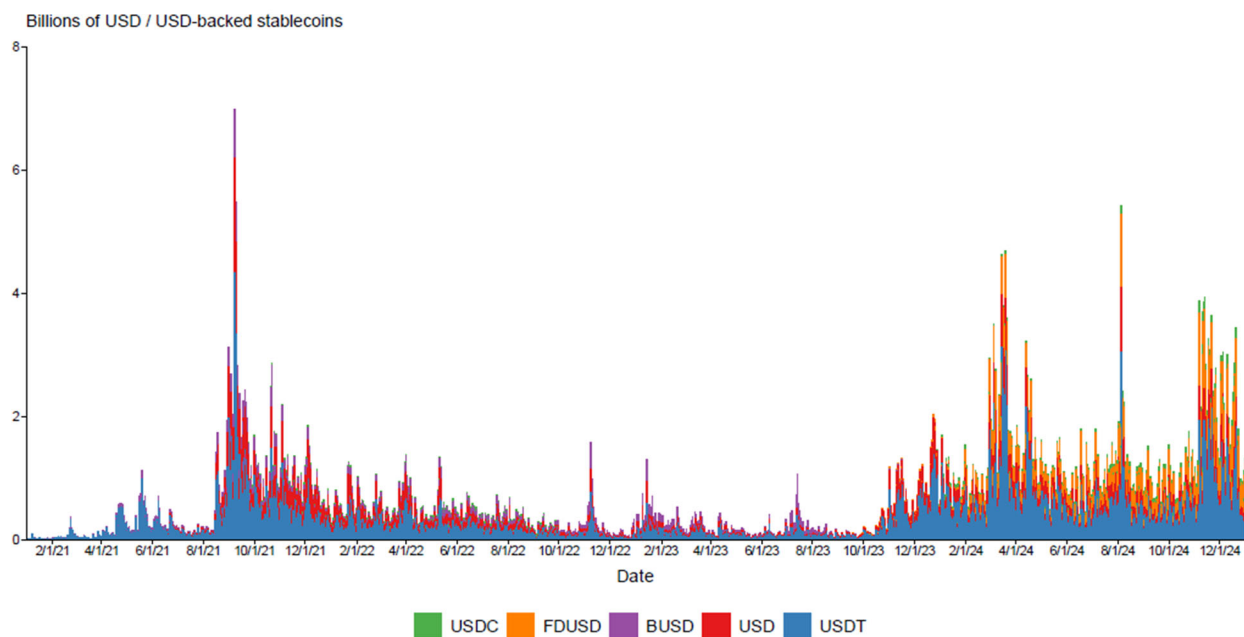
¹¹⁴ See e.g., MacKenzie Sigalos, “From \$10 billion to zero: How a crypto hedge fund collapsed and dragged many investors down with it,” CNBC, July 11, 2022, available at <https://www.cnbc.com/2022/07/11/how-the-fall-of-three-arrows-or-3ac-dragged-down-crypto-investors.html>.

price, and vice versa.¹¹⁵ In 2024, average daily traded volume of SOL perpetual futures exceeded \$3 billion.

65. The SOL token trades against many different cryptocurrencies, including BTC, ETH, USD, and others. The largest volumes are those against USD and USD-backed stablecoins (e.g., USDT and USDC). Figure 2, below, depicts spot SOL traded volume against USD and USD-backed stable coins on the Binance, Coinbase, and Kraken exchanges.

Figure 2

SOL Traded Volume by USD and USD-backed stablecoins, 1/1/21 – 12/31/24



Source: Kaiko

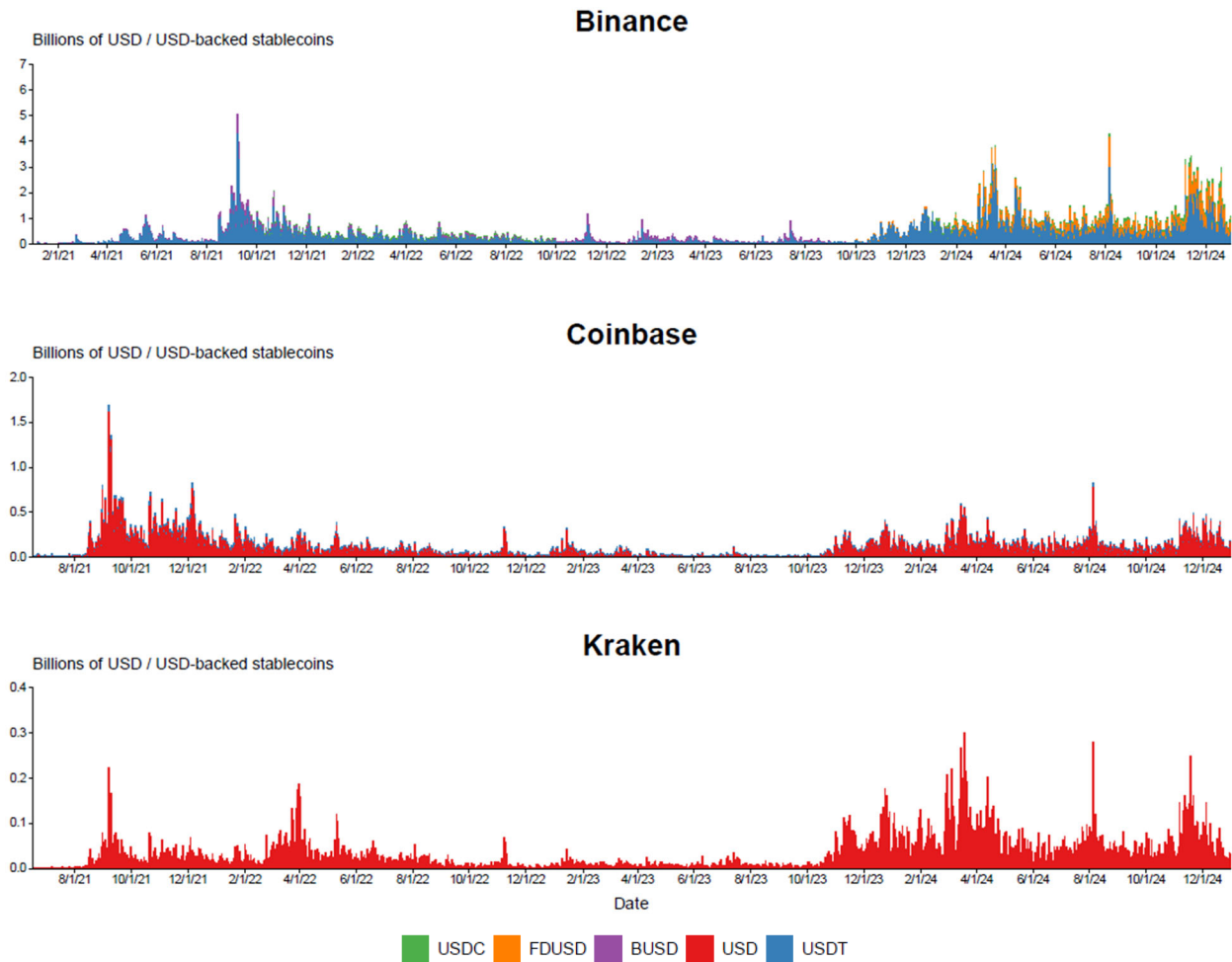
Note: Traded volume includes spot SOL/USDC, SOL/FDUSD, SOL/BUSD, SOL/USD, and SOL/USDT volume on the Binance, Coinbase, and Kraken exchanges. USDC refers to the USD Coin, FDUSD refers to First Digital USD, BUSD refers to Binance USD, and USDT refers to Tether.

66. Globally, Binance has the largest share of SOL trading volume. Within the U.S., Coinbase and Kraken have the largest share of spot volume. Figure 3, below shows SOL traded volume on Binance, Coinbase, and Kraken. This figure shows that SOL trades mostly against USD on Coinbase and Kraken.

¹¹⁵ See e.g., “What are perpetual futures contracts?,” Kraken Lean team, December 2024, available at <https://www.kraken.com/learn/trading/perpetual-futures-contracts>.

Figure 3

SOL Spot Traded Volume by USD and USD-backed stablecoins: Binance, Coinbase, and Kraken, 1/1/24 – 12/31/24



Source: Kaiko

Note: USDC refers to the USD Coin, FDUSD refers to First Digital USD, BUSD refers to Binance USD, and USDT refers to Tether

2. DEX Volumes

67. The analysis of trading and liquidity on DEXs differs from that on CEXs due to the nature of liquidity provision in DeFi applications. Unlike centralized exchanges that rely on order books and market makers, trading on decentralized exchanges is enabled by users depositing tokens into liquidity pools that hold reserves of tokens. This model affects how

liquidity and price impact are measured and evaluated, as they depend on the total value locked in these pools and the AMM algorithms that govern the execution of trades.

68. On DEXs, any user can become a liquidity provider by depositing equal values of each token to a liquidity pool. For example, a user may deposit \$1,000 worth of SOL and \$1,000 worth of USDC stablecoin. The availability of both tokens in a liquidity pool allows other users to exchange their tokens – a user that wishes to acquire SOL would deposit USDC tokens into the pool and receive SOL tokens in exchange. The exchange rate (i.e., price) at which these trades take place is determined by the AMM algorithm.

69. The most basic AMM is a constant product algorithm, which maintains a balance between two token reserves in the pool according to the formula $x * y = k$, where x and y are quantities of two tokens in the pool and k is a constant.¹¹⁶ A trader wishing to acquire one of the tokens from that pool would have to provide a quantity of the other token such that the product of quantities of both tokens in the pool after the trade is executed is still equal to the constant k . Through maintaining this balance, the algorithm explicitly determines the exchange rate.

70. Because both the token balance and AMM constant product formula are known, traders can estimate the price impact of their trades prior to submitting them. In other words, price impact is predictable. This is a key difference between order books (operated by CEXs) and liquidity pools (utilized by DEXs).

71. To understand the concept of price impact in AMM using the constant product formula, we consider a specific example, which we illustrate in Figure 4. Suppose a liquidity pool initially contains $x_1 = 10$ units of cryptocurrency X and $y_1 = 1,000$ units of cryptocurrency Y, with a constant $k = 10 \times 1,000 = 10,000$. At the outset, the price of asset X in terms of asset Y is calculated as $y_1/x_1 = 100$, and the price of asset Y in terms of asset X as $x_1/y_1 = 0.01$. A trader who wishes to purchase 200 units of asset Y will have to supply a certain deterministic quantity of asset X. After the trade, the quantity of asset Y in the pool, i.e., y_2 , will decrease to 800, and to maintain the balance between the assets determined by the constant product $k = 10,000$, the new quantity of asset X will have to be $x_2 = k/y_2 = 10,000/800 = 12.5$. The trader must supply the difference in asset X, and since the initial quantity of asset X in the pool was 10, the trader needs to add 2.5 of asset X to receive 200 of asset Y.

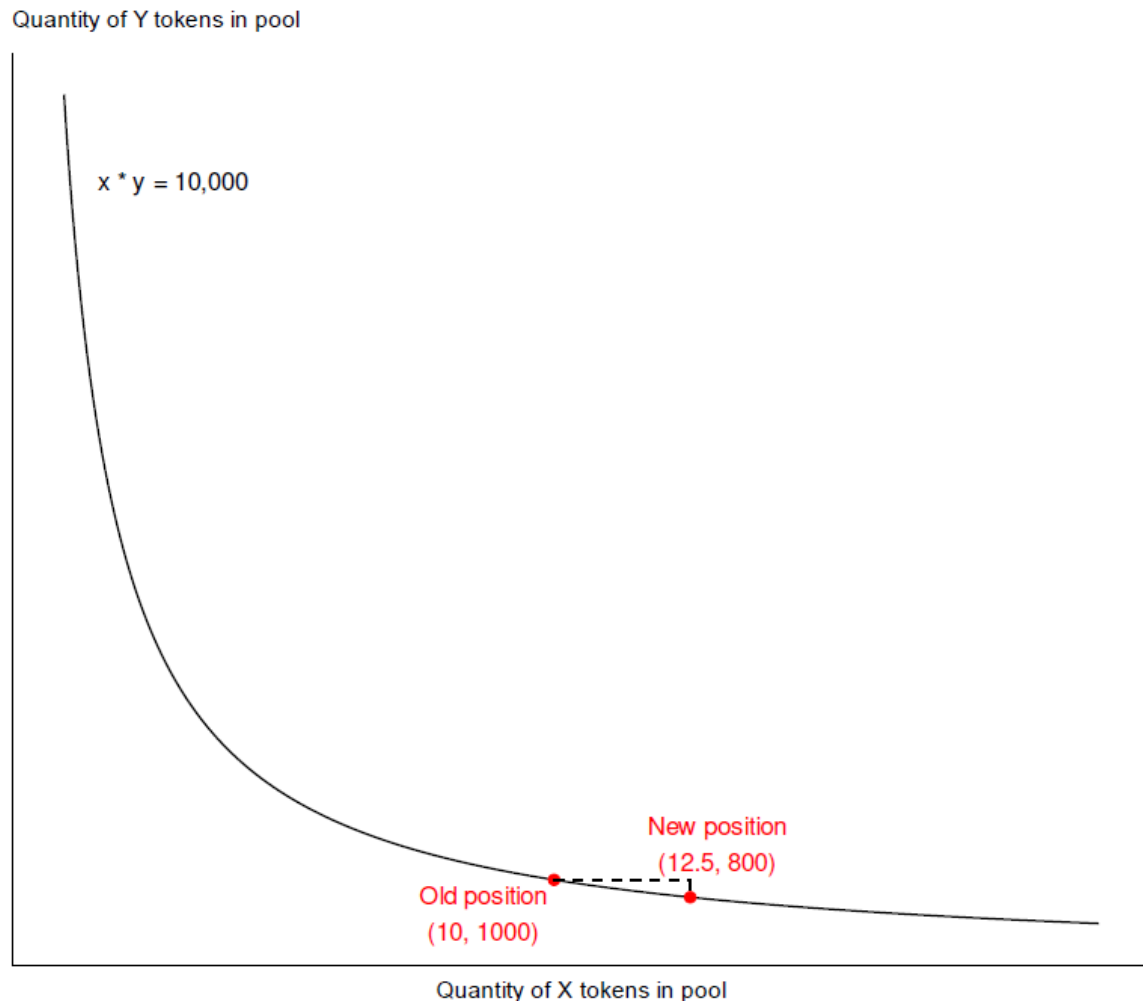
¹¹⁶ See e.g., “What Is an Automated Market Maker,” Gemini, available at <https://www.gemini.com/cryptopedia/amm-what-are-automated-market-makers#section-constant-product-formula>.

72. The price impact in this example becomes evident when examining the price of asset Y before and after the trade. Initially, the price of asset Y was 0.01 (expressed in terms of asset X), and after the trade, it increased to $x_2/y_2 = 12.5/800 = 0.015625$, representing a 56.25% price increase. While the price impact in this example was notable, liquidity pools are typically large, and a single exchange would likely “take out” only a small proportion of tokens from the pool. In particular, if the size of the pool was $x_1 = 1,000$ units of cryptocurrency X and $y_1 = 100,000$ units of cryptocurrency Y, the trader would need to supply approximately 2.004 of asset X to receive 200 of asset Y. The resulting price impact would have been approximately 0.4%. Users of decentralized exchanges see the anticipated price impact in real-time prior to executing a trade, and additionally, typically have a warning displayed if the price impact is unusually high.¹¹⁷

73. Prices of crypto assets are determined by the balances of tokens in liquidity pools and this mechanism allows for arbitrage to keep prices aligned. For instance, a decentralized exchange might offer three liquidity pools (among others): SOL-USDC, SOL-USDT, and USDC-USDT. Suppose a large trade occurs in the SOL-USDC pool, where a trader purchases SOL with USDC, hence increasing the price of SOL in terms of USDC. An arbitrageur can then purchase SOL with USDT, sell SOL for USDC, and exchange USDC back for USDT. In doing so, the arbitrageur would be able to buy SOL at a lower price (in the USDT pool), and sell SOL at a higher price (in the USDC pool), hence obtaining a profit, and as a result, aligning the prices of SOL in both pools. Because transaction fees on Solana are low, such arbitrage opportunities can be captured quickly.

¹¹⁷ See e.g., Uniswap documentation, available at <https://docs.uniswap.org/concepts/protocol/swaps>, (“Approximate price impact is anticipated in real-time via the Uniswap interface, and warnings appear if unusually high price impact will occur during a swap. Anyone executing a swap will have the ability to assess the circumstances of price impact when needed.”).

Figure 4
Stylized Example of Constant Product AMM



74. While the constant product algorithm provides simplicity, it can lead to capital inefficiency and high price impacts as liquidity providers are required to allocate capital uniformly across the entire price range, including price levels that may never be reached.¹¹⁸

Uniswap V3 introduced the concept of concentrated liquidity, which allows liquidity providers to specify price ranges where their capital is utilized.¹¹⁹ Some of the most popular DEXs on Solana, such as Raydium and Orca, employ the AMM algorithms similar to Uniswap V3. Solana's

¹¹⁸ See Adams et al., "Uniswap v3 Core," March 2021, available at <https://app.uniswap.org/whitepaper-v3.pdf>.

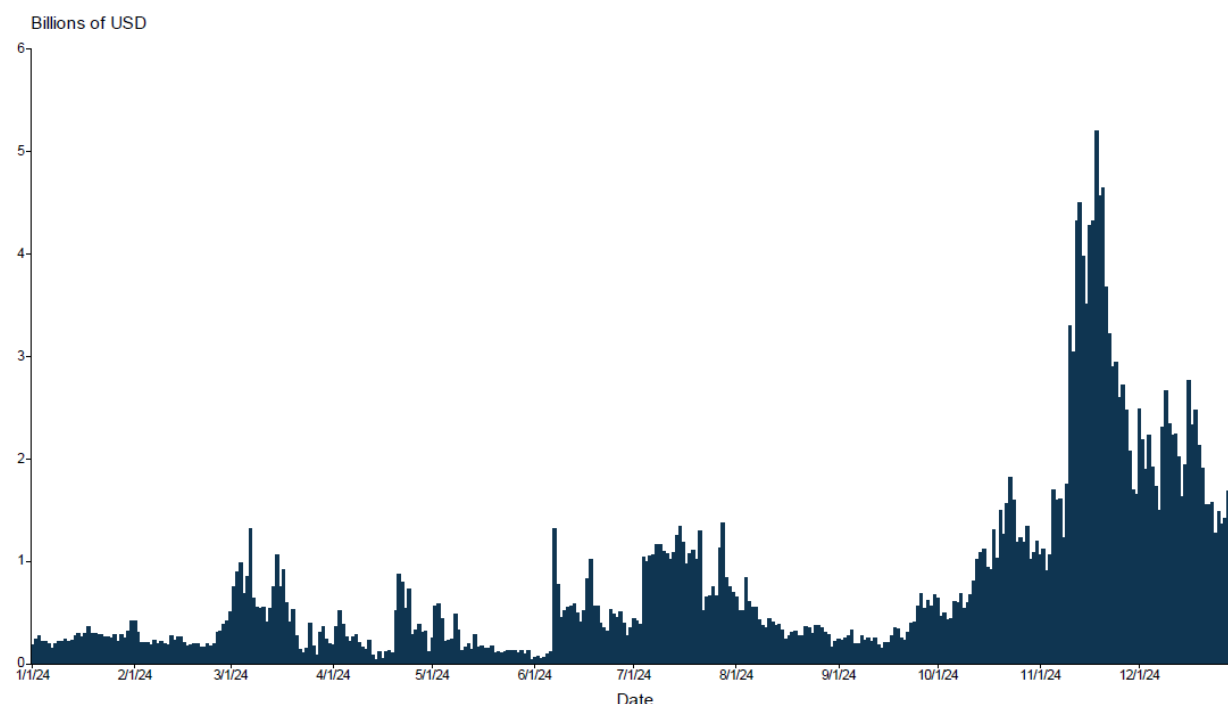
¹¹⁹ Ibid.

architecture also enables the implementation of on-chain Central Limit Order Books (CLOBs), such as OpenBook and Phoenix, with features like real-time order matching.¹²⁰

75. Figure 5, below shows total traded volume on Raydium during 2024. Raydium offers trading in dozens of tokens, most of which trade against SOL. The largest liquidity pools by trade volume are SOL/USDC and SOL/USDT, which have trailing four-week traded volumes totaling more than \$5.2 billion and \$2.5 billion, respectively.¹²¹ Consistent with the uptick in SOL traded volume observed on CEXs, traded volume on Raydium increased during 2024, most notably beginning in October.

Figure 5

Traded Volume on Raydium, January 1, 2024 – December 31, 2024



Source: DeFi Llama

76. The increase in Solana DEX volume has been noted by market observers. For example, Cointelegraph reported that in November 2024, the monthly volume traded on decentralized exchanges operating on Solana surpassed \$100 billion for the first time, reflecting widespread

¹²⁰ See e.g., “A DEX For The People,” OpenBook Home Page, available at <https://www.openbook.ag/>; See also, “The fastest on-chain orderbook in DeFi,” Phoenix Home Page, available at <https://www.phoenix.trade/>.

¹²¹ Volume references are as of January 17, 2025 and reported by Raydium.io.

user participation and the ability of the blockchain to facilitate high economic activity.¹²² The network's active addresses also reached nearly 17 million in that same month, pointing to growing engagement across the blockchain.¹²³ This could be related to Solana's relatively low transaction fees and high throughput. As a reflection of economic activity on Solana, just in November, platforms like Pump.fun and Raydium generated \$71.5 million and \$182 million in fees, respectively.¹²⁴ Solana DEX volume exhibits continued strength into January 2025. Recent reports indicate Solana's DEX volume reached \$27 billion in the one-week period from January 8 to 15, 2025.¹²⁵ This is over \$10 billion more than DEX volume observed on Ethereum.¹²⁶

C. Examining the SOL market: measures of liquidity, market quality, and susceptibility to manipulation

77. The approval of regulated exchange-traded products have been at the forefront of cryptocurrency market discussion for several years. Although not currently available in the U.S., several financial institutions have developed exchange-traded products (ETPs) and funds that provide investors with regulated access to SOL without the need to directly purchase, manage, and safeguard cryptocurrencies. These products allow investors to gain exposure to SOL through traditional financial markets via standard brokerage accounts.

78. Several Solana products are traded on the European exchanges, including 21Shares Solana Staking ETP, CoinShares Physical Staked Solana ETP, ETC Group Physical Solana ETP, and VanEck Solana ETN. In August 2024, two Solana ETFs were approved in Brazil: QR Asset Management Solana ETF and Hashdex Solana ETF.¹²⁷ Figure 6, below shows the value of assets under management held by these products. The total value of SOL held by regulated ETPs and

¹²² Nancy Lubale, "Solana DEX volume hits record high: Is SOL price headed to \$300?," Cointelegraph, November 25, 2024, available at <https://cointelegraph.com/news/solana-dex-volume-record-high-sol-price-to-300>.

¹²³ See, e.g., "Addresses," glassnodestudio, available at <https://studio.glassnode.com/dashboards/onchain-fundamental-metrics?a=SOL>.

¹²⁴ Nancy Lubale, "Solana DEX volume hits record high: Is SOL price headed to \$300?," Cointelegraph, November 25, 2024, available at <https://cointelegraph.com/news/solana-dex-volume-record-high-sol-price-to-300>.

¹²⁵ Oliver Dale, "Solana (SOL) Price Surges Towards \$200 as DEX Volume Surpasses Ethereum," Moneycheck, January 15, 2025, available at <https://moneycheck.com/solana-sol-price-surges-towards-200-as-dex-volume-surpasses-ethereum/>.

¹²⁶ Ibid.

¹²⁷ See e.g., Andrés Engler, "Second Solana ETF Approved in Brazil," CoinDesk, August 21, 2024, available at <https://finance.yahoo.com/news/second-solana-etf-approved-brazil-194007317.html>.

ETFs constitutes approximately 2% of SOL’s market capitalization of \$115 billion (as of 11/29/2024).

Figure 6

Assets Under Management (AUM) of non-U.S. Regulated SOL ETPs

Product	Assets Under Management (AUM)
21Shares Solana Staking ETP	\$1.41 billion (as of 11/27/2024)
CoinShares Physical Staked Solana ETP	\$425.30 million (as of 11/29/2024)
VanEck Solana ETN	\$120.60 million (as of 11/29/2024)
ETC Group Physical Solana ETP	\$26.41 million (as of 11/29/2024)
QR Asset Management Solana ETF	\$9.37 million (as of 11/27/2024)
Hashdex Solana ETF	Not disclosed

Source: 21 Shares; CoinShares; VanEck; ETC Group; QR Asset Management; Hashdex

79. The European ETPs also incorporate staking within their products by delegating the SOL tokens in their custody to validators.¹²⁸ Therefore, an investor who purchases shares of one of these products not only gains exposure to SOL’s price movements but also benefits from staking rewards, which contributes to the nominal amount of SOL tokens represented by each share of these ETPs over time.

80. While U.S. regulators have yet to approve a SOL ETP, the SEC’s approvals of BTC and ETH ETPs in 2024 nonetheless serve as an indication of the substantial developments in cryptocurrency markets and their maturation over the last several years. These approvals also accommodate comparisons between markets for tokens such as SOL and the markets of BTC and ETH.

81. In the subsections that follow, we provide a detailed overview of the SOL market structure and provide insight into the market characteristics potentially relevant to future U.S. regulated product approval. These include key measures of market quality, such as order book liquidity, effective spreads, and trade costs, as well as the degree to which SOL prices are correlated across markets. Each of these measures are important for assessing the market’s

¹²⁸ See e.g., “Bitwise Launches Solana Staking ETP (BSOL) in Europe,” CryptoNinjas, available at <https://www.cryptoninjas.net/news/bitwise-launches-solana-staking-etp-bsol-in-europe/>; See also, “ETP Crypto Staking at 21Shares,” 21Shares, November 2024, available at https://cdn.21shares.com/uploads/current-documents/products/staking/Staking_Primer.pdf.

susceptibility to manipulation. We find that the SOL market is large, liquid, and with characteristics that are broadly comparable to the market for BTC and ETH.

1. Liquidity and Market Quality Measures

82. Liquidity and market quality metrics are foundational concepts for understanding the maturity, efficiency, and stability of any financial market, and this holds true for crypto asset markets. In traditional financial markets, liquidity is typically measured by metrics such as order book depth, bid-ask spreads, and trading volume, all of which reflect the ease with which assets can be purchased and sold without a large price impact.¹²⁹ Market quality is a broader concept that encompasses both liquidity and other empirical measures such as the robustness of price discovery.¹³⁰

83. In the paragraphs that follow, we discuss several liquidity measures and how they have evolved for Solana over time. We compare our SOL metrics to past liquidity estimates for BTC/USD and ETH/USD markets, as found in the literature. This comparative approach helps provide context for Solana's current liquidity profile.

a) Order book depth

84. Order book depth refers to the quantity of buy (bids) and sell (offer) orders at different price levels in a financial market. Essentially, order book depth reflects the supply and demand dynamics for an asset at any given point in time. Order books are divided into two sides: the bid side (buyers) and the offer side (sellers). The highest price at which any individual is willing to purchase is referred to as the "best bid." Conversely, the lowest price at which any individual is willing to sell is referred to as the "best offer." Typically, the best bid price is below the best offer. The midpoint between the best bid and offer is referred to as the "mid-price" or "mid-quote."

85. The depth of the order book is an important indicator of market liquidity. All else equal, markets with deep order books tend to be more liquid, as there are more market participants

¹²⁹ See e.g., Larry Harris, *Trading and Exchanges: Market Microstructure for Practitioners* (New York: Oxford University Press, 2003) ("Harris (2003)"), pp. 68, 70, 394, 398-399.

¹³⁰ See e.g., Maureen O'Hara and Mao Ye, "Is market fragmentation harming market quality?," *Journal of Financial Economics* 100 (2011): pp. 459-474; See also, Harris (2003), pp. 420-421.

willing to buy and sell at levels close to the mid-price, thereby allowing traders to buy and sell assets without having a large price impact and reducing the likelihood of large price swings due to order flow.¹³¹ In contrast, a market with less depth around the mid-price may experience more volatility and larger price impact from large trades. Utilizing historical order-book snapshots captured on Binance, Coinbase, and Kraken, we examine how SOL order-book depth has evolved since its listing on these three CEXs. In our dataset, order book snapshots are sampled roughly every 30 seconds. The exact timing, however, is variable and may be subject to latency. These snapshots capture the quantity of bids and asks at each price level on an exchange's order book.¹³²

86. We begin with Binance, which is the largest exchange by trade volume and one of the most popular CEXs. Binance provides users with the ability to transact in spot, perpetual futures, and other products, and currently lists over 300 cryptocurrencies.

87. Figure 7, below depicts the quantities associated with bids and offers (that is, quoted size) where the bid and offer prices are within 1% and 5% of the mid-price. The dark blue shading below the x-axis quantifies the dollar value of SOL tokens that market participants are willing to buy within one percent of the mid-price. For example, if the mid-price of SOL is \$100, the dark blue shaded region shows the dollar value of bids at prices of \$99 or higher. The light blue shaded region shows how much incremental liquidity is available for bids within 5% of the mid-price. In the example above, the light blue shaded region would show all bids at prices of \$95 and higher.

88. Conversely, the red and yellow shaded region above the x-axis shows the quantity of sell offers at prices within 1% and 5% of the mid-price. Again using the example above, the dark red shaded region would indicate the volume of sell orders market participants have submitted at prices of \$101 and below, and \$105 and below, respectively.

89. As shown in Figure 7, SOL order book depth on Binance has increased substantially over the past two years. In January 2023, the average quantity for bids within 1% of the midpoint was \$1.49 million (\$1.67 million at 5%) and the average quantity for offers within 1% of the

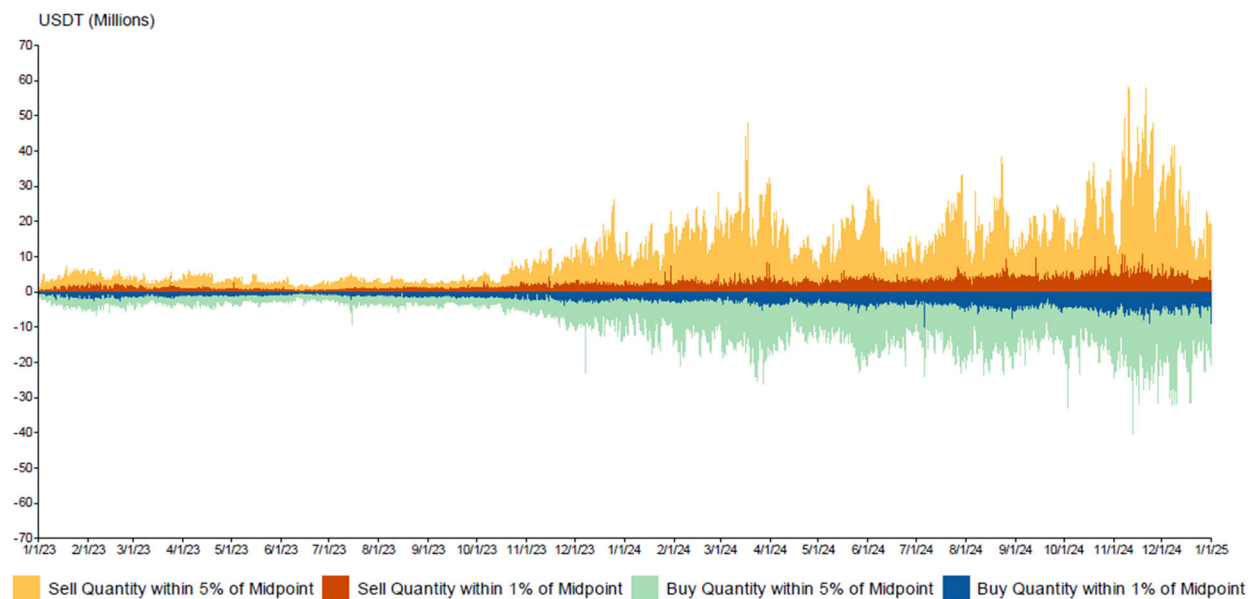
¹³¹ See e.g., Harris (2003), p. 72 ("The price impact of a market order depends on the liquidity available in the market."). Order flow refers to the flow of buy and sell orders entering the market.

¹³² We note that cryptocurrency exchanges, like equity and other exchanges, offer traders the ability to "hide" a portion of the order size from the market. These orders are often referred to as "iceberg" orders. Therefore, to the extent iceberg orders exist in our sample, our measures of liquidity may not include all liquidity available.

midpoint was \$1.46 million (\$2.02 million at 5%). In December 2024, the quantity for bids within 1% and 5% of the midpoint averaged \$5.08 million and \$10.27 million, respectively, and average quantity for offers within 1% and 5% of the midpoint averaged \$4.88 million and \$12.76 million, respectively.

Figure 7

Average Depth around BBO Midpoint by Hour: Binance, 1/1/23 – 12/31/24



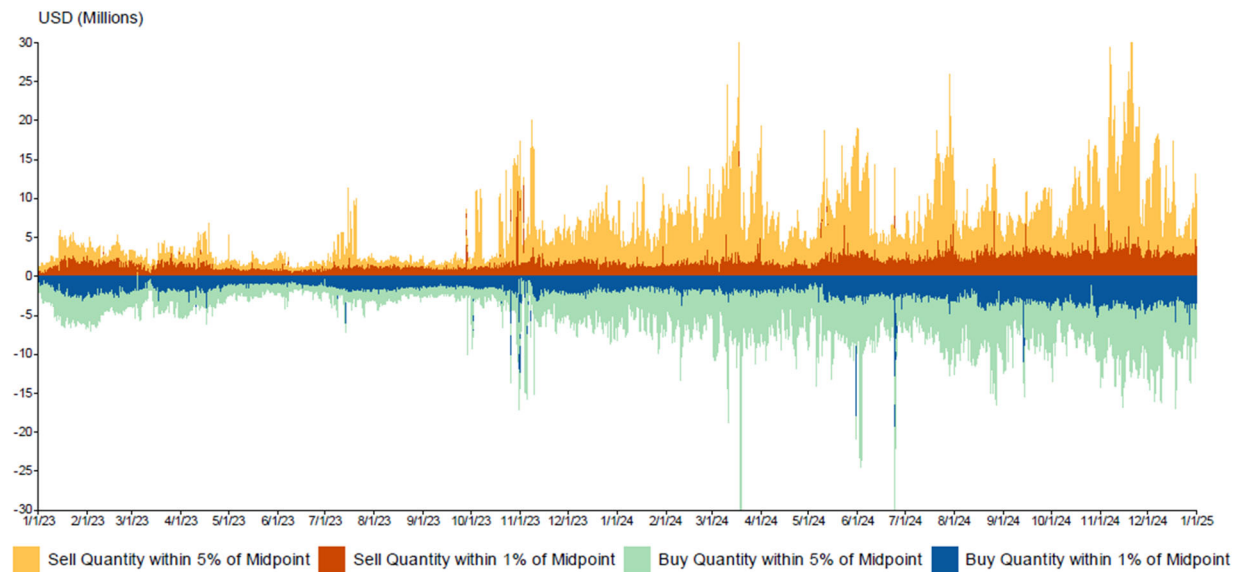
Source: Kaiko

Note: Chart displays the average total buy and sell quantities of SOL/USDT on Binance per hour. Order quantities are grouped by the difference between order price and the midpoint of the best bid and best offer.

90. Figures 8 and 9, below, depict the order-book depth on Coinbase and Kraken, respectively. Similar to the patterns in liquidity observed on Binance, the SOL markets on Coinbase and Kraken have become increasingly liquid. In December 2024, Coinbase averaged more than \$3.73 million in the quantity for bids and \$3.15 million in the quantity for offers within 1% of the midpoint. Kraken averaged \$3.32 million and \$3.47 million.

Figure 8

Average Depth around BBO Midpoint by Hour: Coinbase, 1/1/23 – 12/31/24

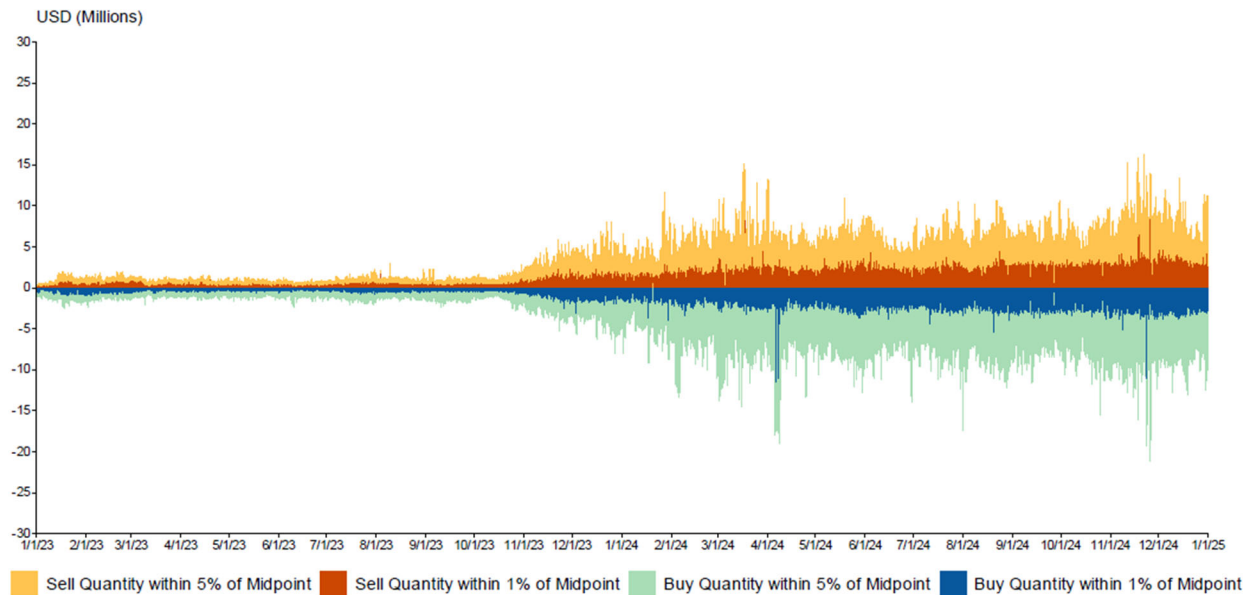


Source: Kaiko

Note: Chart displays the average total buy and sell quantities of SOL/USD on Coinbase per hour. Order quantities are grouped by the difference between order price and the midpoint of the best bid and best offer. Order depth exceeds \$30 million at the 5% level on March 18, March 19, and June 24, 2024.

Figure 9

Average Depth around BBO Midpoint by Hour: Kraken, 1/1/23 – 12/31/24



Source: Kaiko

Note: Chart displays the average total buy and sell quantities of SOL/USD on Kraken per hour. Order quantities are grouped by the difference between order price and the midpoint of the best bid and best offer.

91. SOL liquidity across exchanges shows substantial growth since 2023. Still, it is useful to have a point of comparison to assess SOL liquidity relative to other cryptocurrencies. To do so, we utilize reported data from CoinMarketCap, which captures point-in-time estimates of the quantities associated with bids and offers within 2% of the midpoint across several exchanges and coin-pairs. We compare the largest SOL pairs to the largest BTC and ETH pairs on Binance, Coinbase, and Kraken. As of January 17, 2025:

- The reported depth within 2% of the midpoint of the SOL/USDT market on Binance was \$9.2 million. In contrast, the BTC/USDT market was \$20.5 million and the ETH/USDT market averaged \$14.9 million.
- The reported depth within 2% of the midpoint of the SOL/USD market on Coinbase was \$5.9 million. In contrast, the BTC/USD market was \$24.1 million and the ETH/USD market was \$9.4 million.

- c. The reported depth within 2% of the midpoint of the SOL/USD market on Kraken was \$5.2 million. In contrast, the BTC/USD market was \$22.7 million and the ETH/USD market was \$9.7 million.

92. Comparing order-book depth in USD alone, however, fails to account for the tokens' different market capitalizations and circulating supplies. Both BTC and ETH have substantially larger market capitalizations and token prices relative to SOL. As of January 17, 2025, BTC traded at ~\$104,400 per token and had a market capitalization of \$2.06 trillion; ETH traded at ~\$3,450 per token and had a market capitalization of \$416 billion. In contrast, SOL traded at ~\$220 and had a market capitalization of \$108 billion. Moreover, the circulating supplies of BTC (19.8 million), ETH (120.5 million), and SOL (486.6 million) vary considerably.

93. Normalizing order-book depth to account for the percentage of circulating coins (rather than USD value) provides another measure of liquidity. Figure 10, below, presents these values for SOL, BTC, and ETH. As Figure 10 shows, a larger percentage of the SOL circulating supply is available at prices close to the midpoint. For example, over 42,000 SOL tokens are available to be traded within 2% of the midpoint price on Binance. This represents approximately 0.008685% of the circulating supply. In contrast, 196 BTC coins are available to trade within 2% of the midpoint, representing 0.000991% of the BTC circulating supply.

Figure 10

Liquidity comparison across SOL, BTC, and ETH, January 17, 2025

Metric	BTC	ETH	SOL
Price	\$104,433	\$3,462	\$218
Circulating supply	19,810,000	120,500,000	486,590,000
<i>Binance</i>			
Liquidity +/- 2%			
USD	\$20,500,000	\$14,900,000	\$9,200,000
Coins	196	4,304	42,262
Percentage of Circulating Supply +/- 2%	0.000991%	0.003572%	0.008685%
<i>Coinbase</i>			
Liquidity +/- 2%			
USD	\$24,100,000	\$9,400,000	\$5,900,000
Coins	231	2,715	27,103
Percentage of Circulating Supply +/- 2%	0.001165%	0.002253%	0.005570%
<i>Kraken</i>			
Liquidity +/- 2%			
USD	\$22,700,000	\$9,700,000	\$5,200,000
Coins	217	2,802	23,887
Percentage of Circulating Supply +/- 2%	0.001097%	0.002325%	0.004909%

Source: CoinMarketCap

b) Market quality metrics and liquidity measures

94. While order-book depth is an important component to assessing market quality and liquidity, it is by no means the only one. Measuring liquidity has been the focus of a substantial portion of the academic finance literature and receives considerable attention from researchers, regulators, and market participants alike.

95. Liquidity is generally defined as having the ability to trade large volumes quickly with minimal price impact. In liquid markets, market participants can rapidly trade on their information with minimal transaction costs. Moreover, liquid markets present greater barriers for would-be market manipulators, as they must deploy substantial capital to achieve even modest price movements, thereby reducing the feasibility (and profitability) of price manipulation. The opposite is true in thinly traded, or “illiquid” markets. In markets with low liquidity, even moderately-sized trades by someone attempting to manipulate can exert price

pressure,¹³³ potentially leading to temporary or even a prolonged mispricing, especially if it is too costly for arbitrageurs to correct such a mispricing.

96. Utilizing intraday data obtained from Kaiko, we estimate four high-frequency liquidity measures that capture insights into multiple facets of the liquidity, resiliency, and the general market quality of the SOL token: (i) percentage quoted spread, (ii) percentage effective spread, (iii) percentage price impact, and (iv) cost of a roundtrip trade. These measures show how easily, quickly, and inexpensively traders can transact, as well as how much the price deviates in response to trades. They are widely used in the academic literature. By examining these metrics collectively, we can assess both the liquidity and resiliency of the SOL token markets, providing a nuanced understanding of its overall market quality.

(1) Percentage quoted spread

97. The percentage quoted spread (“PQS”) measures how close the best bids and asks are from the midpoint. It is calculated by taking the difference between the best ask and the best bid and expressing that difference as a percentage of the midpoint price. For example, if an asset has a best bid of \$99 and a best ask of \$101, then percentage quoted spread is the two dollar difference between the best bid and ask, expressed as a percentage of the midpoint price of \$100. In this example, the percentage quoted spread is 2%. Quoted spreads matter because they provide a quick snapshot of the cost to trade immediately. All else equal, lower percentage spreads indicate buyers and sellers are willing to trade at prices close to the midpoint, and traders can, therefore, have a smaller price impact (up to the quantity at the best bid/ask).

98. Utilizing historical order-book snapshots, we calculate the PQS as follows. For each orderbook snapshot, we identify the best bid and best ask prices.¹³⁴ We then scale the spread (best ask less best bid as explained above) by the mid-point of the two to get a point-in-time estimate for the PQS. To reduce noise and provide a more stable measure, we first average at the daily level to get the average daily PQS. To further smooth out daily fluctuations, we then compute 90-day rolling averages of the daily PQS values.

¹³³ See e.g., Harris (2003), p. 72.

¹³⁴ As noted above, our data contains roughly two orderbook snapshots every minute.

99. Figure 11 plots the percentage quoted spread of SOL on Binance, Coinbase, and Kraken from April 2023 through December 2024. We note two important findings. First, the average percentage quoted spread has been under 10 basis points on Coinbase and under 5 basis points on Binance and Kraken for at least the past year. Second, between 2023 and 2024, average percentage quoted spreads declined substantially. This is consistent with a market that has matured, as evidenced by the increase in order-book depth and trade volume described above.

100. Average spreads from the last three months of 2024 were the lowest over the period, ranging from 0.53 basis points on Binance to 0.78 basis points on Coinbase. These are comparable to and in some cases better than spreads estimated by the literature studying past liquidity in crypto markets. For example, Brauneis et al. (2021) estimate several high- and low-frequency liquidity measures for the BTC/USD and ETH/USD markets by compiling a high-frequency dataset that contains transactional data from the two-year period from 12/16/2017 to 12/16/2019 across three of the then-largest trading venues: Bitfinex, Bitstamp, and Coinbase Pro.¹³⁵ The authors construct high-frequency measures of transaction costs and price impact, which “serve as [their] liquidity benchmarks.”¹³⁶ Brauneis et al. (2021) find average quoted spreads for the BTC/USD market ranged from 0.636 basis points on Coinbase Pro to 6.55 basis points on Bitstamp.¹³⁷ For the ETH/USD market ranging from 1.232 basis points on Coinbase Pro to 13.17 basis points on Bitstamp.¹³⁸ While these figures are not directly comparable due to differing analysis periods, we note that the market capitalization of BTC at the end of the authors’ analysis period (12/16/2019) was approximately \$130 billion. The market capitalization of SOL at the end of our analysis period (\$93 billion as of 12/31/2024),¹³⁹ reflects a similar scale.

¹³⁵ Alexander Brauneis et al., “How to Measure the liquidity of cryptocurrency markets?,” *Journal of Banking and Finance* 124 (2021) (“Brauneis et al. (2021)”), p. 2 (“We use a novel and comprehensive set of continuous transactions data and order book snapshots comprising the 50 best bids and asks for two major cryptocurrencies (bitcoin and ethereum) and three large exchanges (Bitfinex, Bitstamp and Coinbase Pro) over a two-year period.”).

¹³⁶ Brauneis (2021), p. 2 (“First, we use these data to construct high-frequency measures of transaction costs and price impact. These measures serve as our liquidity benchmarks.”).

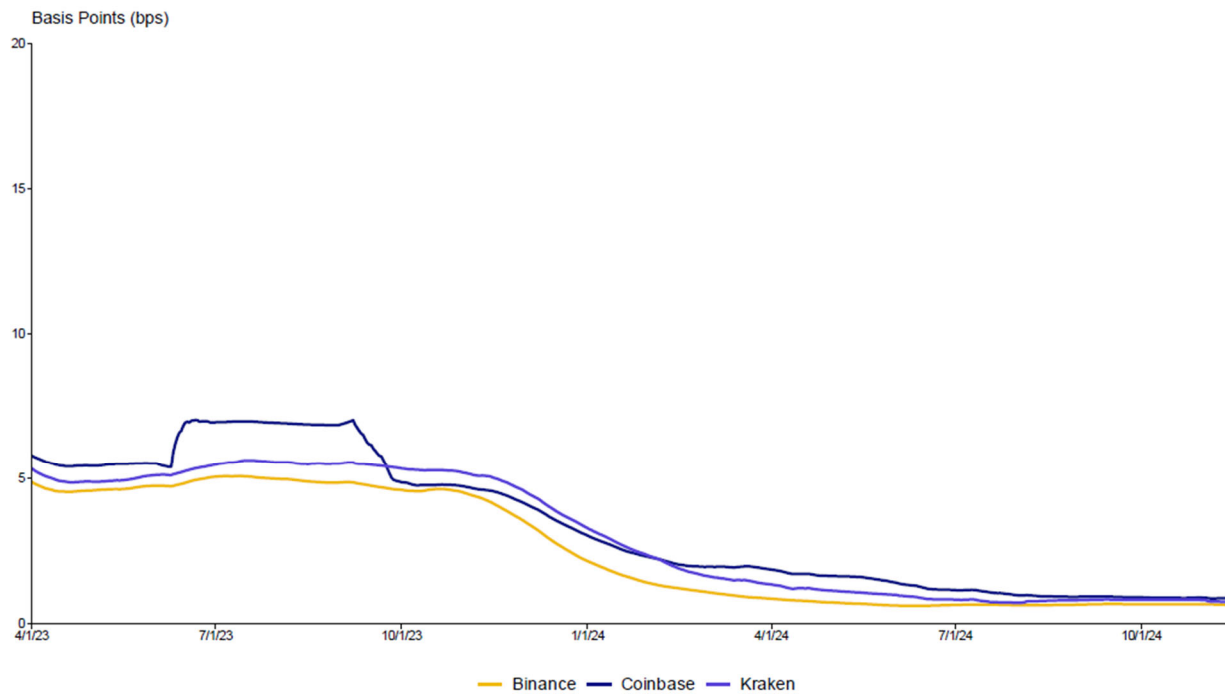
¹³⁷ See Brauneis (2021), Table 3.

¹³⁸ See Brauneis (2021), Table 9.

¹³⁹ See e.g., CoinMarketCap.

Figure 11

Average Percentage Quoted Spread: 90-Day Rolling Window, 4/1/23 – 12/31/24



Source: Kaiko

Note: Chart displays the average percentage quoted spread for the following coin pairs: SOL/USDT on Binance, SOL/USD on Coinbase, and SOL/USD on Kraken.

(2) Percentage effective spread

101. Whereas the quoted spread measures the spreads observed in the market *prior* to a transaction, the effective spread measures the realized spread *after* a transaction is consummated. Effective spreads reflect the difference between the traded price of an asset and the mid-price observed immediately prior to the transaction. Effective spreads therefore take into account the degree to which a large transaction “walks the book,” as well as any liquidity that was hidden from the market (e.g., iceberg orders) and hence reflect a measure closer to the actual transaction costs faced by a trader, rather than the cost implied by the quoted spreads. Effective spreads are

generally considered to be more realistic indicator of market liquidity than quoted spreads.¹⁴⁰

This is especially true because, as mentioned earlier, the actual trades can occur both inside the quotes (if there's hidden liquidity) or even partly outside the quotes if the liquidity at the top of the book is not sufficient. Effective spreads also account for the possibility that the traders receive price improvement, i.e., buyers executing their trades below the ask or sellers executing orders above the bid. One of the most common ways to express effective spreads is as a percentage of the mid-price. Expressing the effective spread as a percentage of the mid-price also allows for a direct comparison to the percentage quoted spreads described above and controls for differences in asset price levels.

102. We estimate the percentage effective spread of the SOL token on Binance, Coinbase, and Kraken using the following formula:

$$ES_t = \frac{1}{N_t^+} \sum_{j=1}^{N_t^+} \frac{2|P_j^+ - MQ_j|}{MQ_j}$$

where P_j^+ is the price of first transaction that occurs after the orderbook snapshot, N_t^+ is the number of order book snapshots that are followed by a transaction before the next order book snapshot is recorded,¹⁴¹ and MQ_j is the mid-point of bid and ask prices.

103. For example, building on our percentage quoted spread example, suppose an asset has a best bid of \$99 and a best ask of \$101, and the transaction occurs at \$100.5. Then, the percentage effective spread is twice the absolute value of the half-dollar difference between the executed price and the mid-quote (\$100), expressed as a percentage of the mid-quote of \$100. In this example, the percentage effective spread is 1%.

104. Figure 11, below plots the average effective spreads of SOL on Binance, Coinbase, and Kraken from April 2023 through December 2024. Similar to the trends observed in Figure 10, the average percentage effective spreads have declined over time. Importantly, the average effective spreads are largely comparable to the quoted spreads, indicating that observed liquidity in the market is tradeable in sufficient quantity.¹⁴²

¹⁴⁰ See e.g., Mitchell A. Petersen and David Fialkowski, "Posted versus effective spreads: Good prices or bad quotes?," *Journal of Financial Economics* 35, no. 3 (1994): pp. 269–292.

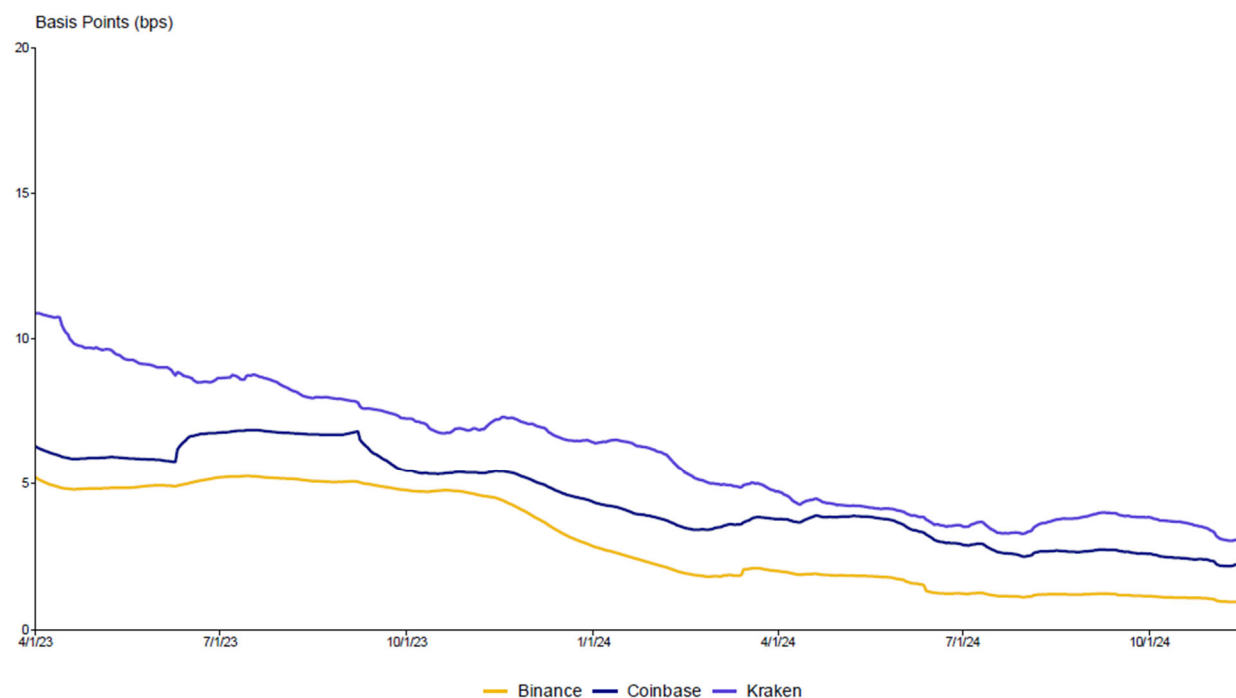
¹⁴¹ If there are no transactions between two orderbook snapshots, then the first snapshot is not used. If there are multiple transactions between two orderbook snapshots, then we take the price of the first transaction.

¹⁴² See Figure 14, below.

105. As observed with the average quoted spreads, average effective spreads were the lowest over the last three months of 2024, ranging from 0.95 basis points on Binance to 3.13 basis points on Kraken. As a point of comparison Brauneis et al. (2021) report effective spreads in BTC/USD ranging from 1.04 – 6.99 basis points and in ETH/USD ranging from 1.929 – 13.45 basis points.¹⁴³

Figure 12

Average Percentage Effective Spread: 90-day Rolling Window, 4/1/23 – 12/31/24



Source: Kaiko

Note: Chart displays the average percentage effective spread for the following coin pairs: SOL/USDT on Binance, SOL/USD on Coinbase, and SOL/USD on Kraken.

(3) Percentage price impact

106. Price impact is yet another way to estimate liquidity in the market. Instead of measuring the distance between bids and asks (as in the quoted spread) or the difference between traded price and the mid-price (as in effective spread), price impact measures the change in the mid-

¹⁴³ See Brauneis (2021), Table 3; Table 9.

price before and after a trade. The price impact measure reflects an estimate of how much a trade impacts the price, and thus measures the asset's liquidity.¹⁴⁴ Price impact estimates can isolate the component of price changes directly attributable to trading activity, thereby facilitating cross-sectional and time-series comparisons of market depth.¹⁴⁵ We express the price impact as a percentage of the mid-price, just as we did in the spread measures described above. All else equal, a higher percentage price impact signifies lower liquidity and vice versa. Even in markets with tight quoted spreads, notable price movements can occur if the order book is shallow, i.e., available quantity to trade is low. This is because quoted spreads primarily reflect the cost of executing typical (small to average-sized trades), and may not capture larger than average trades.¹⁴⁶

107. Price impact measures can account for the effects of larger trades, which are more likely to cause substantial price movements. While effective spreads capture some of this information by reflecting actual execution costs, they are limited to capturing the immediate and potentially temporary price shift. In contrast, price impact can capture more permanent price movements, which can be driven by the arrival of new information. Finally, price impact measures how demand reveals itself through a *persistent* shift in prices unlike measures of spread, which capture the *immediacy* of trading. Therefore, a high price impact can indicate lower market *resilience*, which can signal heightened vulnerability to volatility.

108. We estimate the percentage price impact of the SOL token on Binance, Coinbase, and Kraken using the following formula:

$$PI_t = \frac{1}{N_t^+ - 1} \sum_{j=1}^{N_t^+ - 1} \frac{Q_j^+ \cdot (MQ_{j+1} - MQ_j)}{MQ_j}$$

¹⁴⁴ This type of price impact can also be referred to as “slippage” or “implementation shortfall” and has attracted much attention from researchers, both in terms of how to measure it and how to manage it.

¹⁴⁵ See e.g., Joel Hasbrouck, “Measuring the Information Content of Stock Trades”, *The Journal of Finance* 46, no. 1 (1991): pp. 179-207.

¹⁴⁵ Trade indicator is defined by the net trades volume. That is, it is +1 if the buy value of trades exceeds the sell value and -1, otherwise.

¹⁴⁶ Average or typical trade size can vary across asset classes or within an asset class (e.g., small cap vs large cap equities).

where Q_j^+ denotes the trade indicator (+1 for a buyer-initiated trade and -1 for a seller-initiated trade),¹⁴⁷ MQ_j is the mid-point of bid and ask prices, and N_t^+ , is the number of order book snapshots that are followed by a transaction before the next order book snapshot is recorded.

109. Because our order book data is limited (two snapshots per minute), our measure differs from those typically calculated using high-frequency data. In particular, we calculate *net* trade volume between snapshots to inform our Q_j^+ , rather than individual trades. As a result, our measures are not directly comparable to those reported by Bruneis et al. (2021).¹⁴⁸ In effect, our measure captures the price impact of *net* trading between snapshots, rather than the price impact of any *individual* trade.

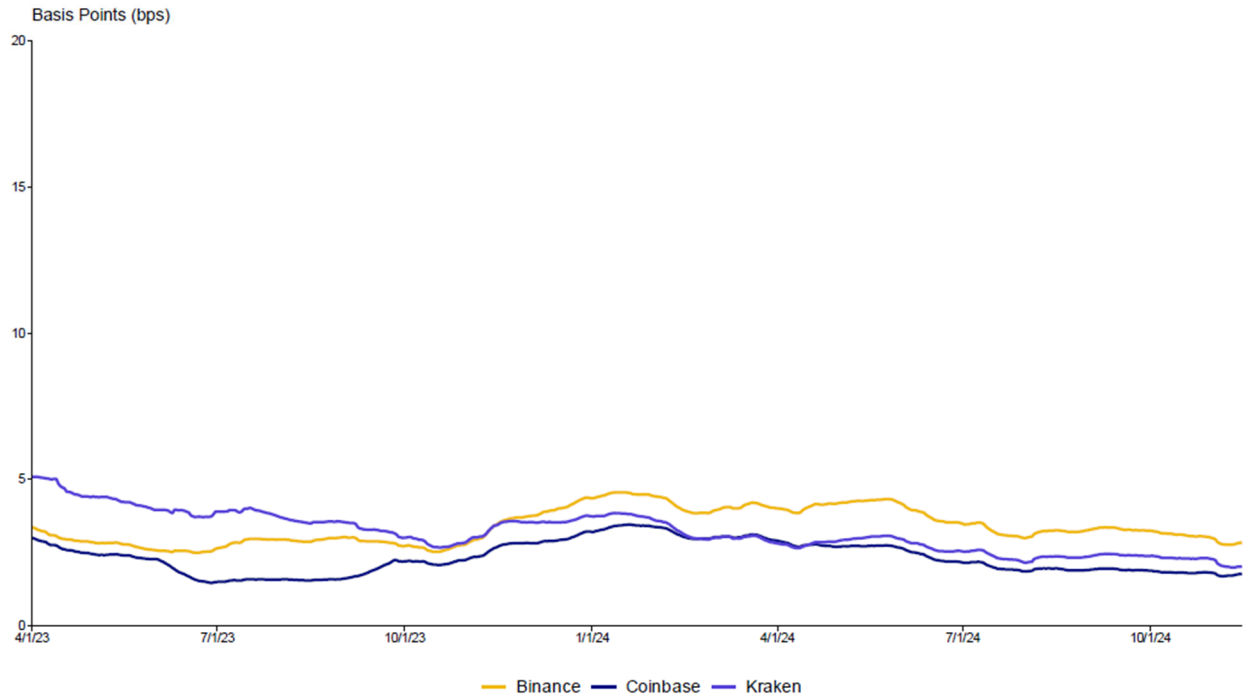
110. Figure 12, below plots the average price impact of SOL on Binance, Coinbase, and Kraken from April 2023 through December 2024. We note that the average percentage price impact has been relatively stable and approximately five basis points or less. These numbers are comparable to our spread measures (*see* Figure 14) and further indicate that the SOL market has become increasingly liquid. Interestingly, price impact estimates are largest on Binance, even though Binance had the lowest spread measures among the three exchanges. This could be a result of the relatively higher trade volume on Binance, which directly affects how our price impact measure is estimated (using net trade volume between snapshots). This may also be indicative of price discovery occurring on the Binance exchange.

¹⁴⁷ The transaction data provided by Kaiko (and recorded from exchanges) indicates whether a trade is buyer or seller initiated. Due to the frequency at which our orderbook snapshots are sampled (two per minute), we use the net trade volume to inform our estimation of Q_j^+ .

¹⁴⁸ *See* Bruneis et al. (2021), p. 5 (“To estimate the price impact we use data sequences consisting of an order book snapshot, the first transaction after the snapshot and the subsequent order book snapshot.”).

Figure 13

Average Price Impact: 90-Day Rolling Window, 4/1/23 – 12/31/24



Source: Kaiko

Note: Chart displays the average percentage price impact for the following coin pairs: SOL/USDT on Binance, SOL/USD on Coinbase, and SOL/USD on Kraken.

(4) Percentage cost of a roundtrip trade

111. Lastly, we provide a measure of the SOL market's resiliency to large transaction volume by analyzing the percentage cost of a roundtrip trade. We compute the percentage cost of a roundtrip trade by simulating the cost of simultaneously buying and then selling SOL (or vice-versa) for a trade size in the 99.9th percentile of distribution, scaled by the mid-quote, thereby capturing the total estimated transaction cost of a particularly *large* order. This measure provides insight into market depth and liquidity resilience under extreme order flow demands.

We calculate percentage cost of a roundtrip trade (CRT (Y)) as follows:

112. First, we estimate the weighted average price of executing a transaction of size Y using the following formula:

$$P(Y) = \frac{\sum_{k=1}^K A_k \cdot V_k}{\sum_{k=1}^K V_k}$$

113. The formula above is subject to the condition that $\sum_{k=1}^K A_k \cdot V_k = Y$. A_k is the price of k^{th} order and V_k is the volume of k^{th} order. Y equals the 99.9% quantile of the trade size distribution. In our data, the 99.9% quantile across exchanges, rounded to the nearest dollar, is \$25,572.

114. To estimate the cost of a roundtrip trade of a certain size (Y), we calculate the average prices for buying and selling that size at market rates, find the difference between the two prices and express it as a percentage of the midpoint, and then take an average of these calculations across all snapshots of the orderbook during a given time interval (t). This gives us the estimated cost of a roundtrip trade, denoted as $\text{CRT}(Y)$.

115. There are several advantages to examining this metric. First, large trades are more likely to “stress-test” the liquidity of the market and this measure provides an estimate for the transactions costs associated with such orders. Unlike smaller sized trades, large orders are likely to have a notable effect on price, and this measure specifically estimates price slippage and costs incurred when executing large trades (assuming liquidity in the book does not change). Second, this measure provides insights into the “tail end” of liquidity distribution, which can be crucial in understanding how well a market may accommodate extreme trading events.

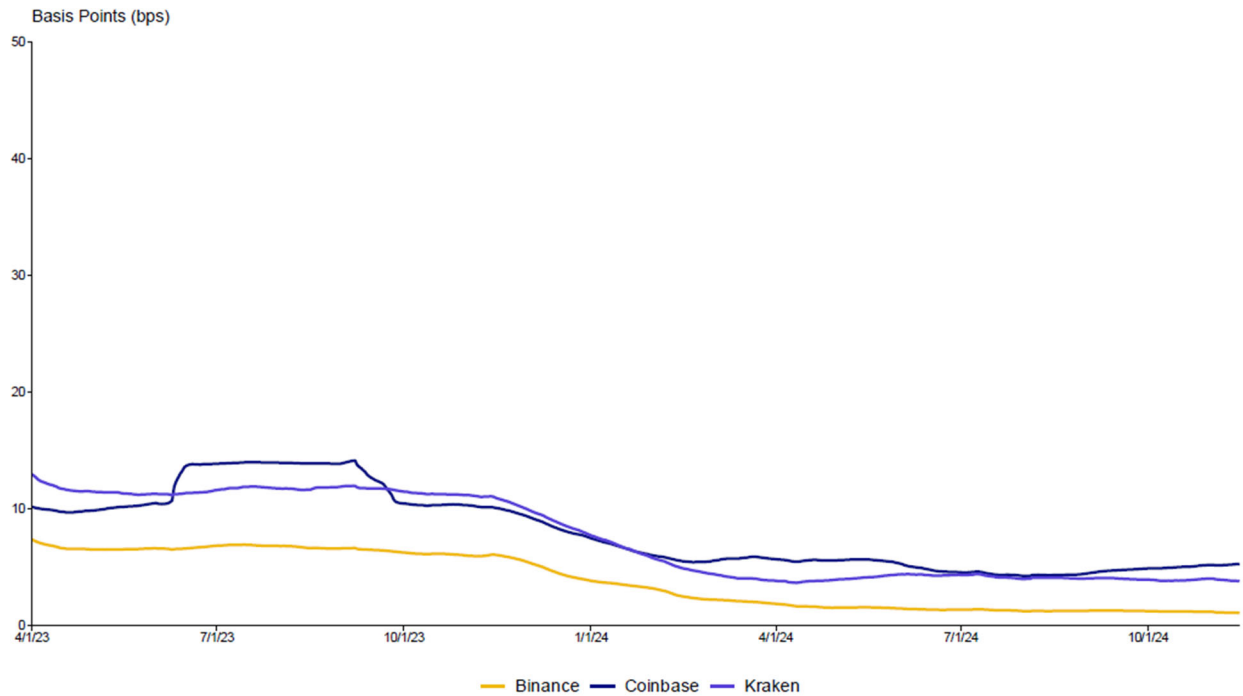
116. Figure 14 below plots the average percentage cost of a roundtrip trade of SOL on Binance, Coinbase, and Kraken from April 2023 through December 2024. As shown in Figure 14, the average percentage cost of a roundtrip was below 10 basis points across all exchanges in 2024. As with the quoted and effective spreads, the cost of a roundtrip trade is lowest in the last 90 days of 2024, ranging from 1.02 basis points on Binance to 5.02 basis points on Coinbase. These are broadly comparable to BTCUSD metrics found by Bruneis et al. (2021), which ranged from 3.41 – 13.23 basis points,¹⁴⁹ and better than ETHUSD metrics, which ranged from 7.22 – 30.33 basis points.¹⁵⁰

¹⁴⁹ See Bruneis et al. (2021), Table 3.

¹⁵⁰ See Bruneis et al. (2021), Table 9.

Figure 14

Average Percentage Round-trip Trade Cost: 90-Day Rolling Window, 4/1/23 – 12/31/24



Source: Kaiko

Note: Chart displays the average percentage round-trip trade cost for the following coin pairs: SOL/USDT on Binance, SOL/USD on Coinbase, and SOL/USD on Kraken.

117. In sum, the SOL market shows robust liquidity. Figure 15, below provides a summary of the overall distribution of the liquidity measures discussed above for the period of October 1, 2024 to December 31, 2024. Across all exchanges, the evidence suggests SOL markets have been liquid, have had low transaction costs, and have been generally resilient to shocks from large-sized orders.

Figure 15

Liquidity of SOL on Binance, Coinbase, and Kraken, 10/1/24 – 12/31/24

Quote Spread						
Exchange	Q1	Median	Q3	Mean	Standard Deviation	Count
Binance	0.44	0.51	0.61	0.53	0.12	312,943
Coinbase	0.46	0.59	0.89	0.78	0.55	317,725
Kraken	0.45	0.55	0.67	0.68	0.64	264,910

Effective Spread						
Exchange	Q1	Median	Q3	Mean	Standard Deviation	Count
Binance	0.45	0.56	0.69	0.95	1.32	312,942
Coinbase	0.57	1.29	3.20	2.50	4.23	317,648
Kraken	0.50	0.69	3.99	3.13	4.84	226,185

Price Impact						
Exchange	Q1	Median	Q3	Mean	Standard Deviation	Count
Binance	0.00	2.39	6.06	3.20	6.81	312,941
Coinbase	-1.48	1.40	5.16	2.03	7.37	317,647
Kraken	-0.70	0.45	4.81	1.97	7.57	226,184

Round Trip Cost						
Exchange	Q1	Median	Q3	Mean	Standard Deviation	Count
Binance	0.60	0.85	1.27	1.02	0.60	312,943
Coinbase	3.94	4.90	5.92	5.02	1.84	317,725
Kraken	1.90	2.81	4.74	3.58	2.59	264,910

Source: Kaiko

Note: All liquidity measures are presented in basis points.

2. SOL returns are highly correlated across exchanges

118. Another metric relevant for assessing SOL's market quality is the degree to which SOL spot prices are consistent across exchanges. In a well-functioning market, we would expect SOL prices across exchanges to be highly correlated due to two characteristics inherent to the market. First, SOL tokens are fungible. As is the case with a dollar bill, a share of stock, or a bar of gold, each SOL token is interchangeable with the next—buyers do not care which SOL token they receive and sellers do not care which SOL token they sell. Second, market participants can and do trade on multiple exchanges. Moreover, the Solana network's fast transaction speeds and low costs enable quick transfers of tokens between users and exchanges, making the transfer of tokens from one exchange to another uncomplicated and inexpensive. Together, these features provide a mechanism for arbitrageurs to quickly exploit mispricing and reduce attempts to manipulate SOL prices on a single exchange.

119. To empirically test the correlation between SOL markets, we again draw upon the price and transaction data acquired from Kaiko. To evaluate cross-market correlation, our analysis

examines the time series of SOL returns observed on Binance, Coinbase, and Kraken. From the transaction data, we construct returns at one-minute, five-minute, and one-hour levels for the SOL/USDT and SOL/USD trading pairs.¹⁵¹ Two types of correlation statistics are calculated: (i) the first uses the entire sample period, and (ii) the second estimates correlation using a rolling 90-day window over the same sample period. Figures 16 and 17, below, report the results.

Figure 16
Correlation of SOL Returns on Select Centralized Exchanges (Entire Sample 6/19/21 – 11/14/24)

one-minute window		
Exchange	Binance	Coinbase
Coinbase	0.948	—
Kraken	0.825	0.820

five-minute window		
Exchange	Binance	Coinbase
Coinbase	0.983	—
Kraken	0.933	0.931

one-hour window		
Exchange	Binance	Coinbase
Coinbase	0.998	—
Kraken	0.990	0.990

Source: Kaiko.

Note: Returns are calculated using the last price at each one-minute, five-minute, or one-hour interval. The correlation measure used is Pearson's correlation coefficient. The correlation between two exchanges is calculated for timestamps for which the returns are not null for both exchanges. The first date used in the correlation calculations is 6/19/21, which is the first date where returns are available for all products shown. The following products are used: SOL-USDT on Binance; and SOL-USD on Coinbase and Kraken.

¹⁵¹ The SOL/USD pair is used for Kraken and Coinbase as it represents the largest share of SOL trading volume.

Figure 17***Correlation of SOL Returns on Select Centralized Exchanges (Rolling 90-Day Window on the Sample 6/19/2021 – 11/14/2024)***

one-minute window (min / max)		
Exchange	Binance	Coinbase
Coinbase	0.856 / 0.99	—
Kraken	0.755 / 0.888	0.749 / 0.88

five-minute window (min / max)		
Exchange	Binance	Coinbase
Coinbase	0.936 / 0.997	—
Kraken	0.856 / 0.973	0.86 / 0.971

one-hour window (min / max)		
Exchange	Binance	Coinbase
Coinbase	0.991 / 1.000	—
Kraken	0.971 / 0.998	0.972 / 0.998

Source: Kaiko

Note: Returns are calculated using the last price at each one-minute, five-minute, or one-hour interval. The correlation measure used is Pearson's correlation coefficient. The correlation between two exchanges is calculated for timestamps for which the returns are not null for both exchanges. The first date used in the correlation calculations is 6/19/21, which is the first date where returns are available for all products shown. The following products are used: SOL-USDT on Binance; and SOL-USD on Coinbase and Kraken.

120. The correlation coefficients in Figures 16 and 17 all indicate a strong positive correlation between SOL returns across exchanges. Importantly, correlations at longer intervals are higher than correlations at shorter intervals. For example, hourly returns are more closely aligned across exchanges than SOL returns at the minute level. This indicates that temporary price dislocations – perhaps caused by order flow or liquidity changes – dissipate within the hour, suggesting the existence of an arbitrage mechanism that can exploit temporary price discrepancies.

121. Susceptibility to manipulation has long been at the forefront of U.S. regulators concerns surrounding the approval of cryptocurrency ETPs. In its Bitcoin spot ETF's approval, the SEC noted that "one way an exchange that lists [B]itcoin-based exchange-traded products ('ETPs') can meet the obligation under Exchange ACT Section 6(b)(5) that its rules be designed to prevent fraudulent and manipulative acts and practices is by demonstrating that the exchange has a comprehensive surveillance-sharing agreement with a regulated market of significant size

related to the underlying or reference [B]itcoin assets.”¹⁵² It also further explained that “[A] listing exchange could, alternatively, demonstrate that ‘other means to prevent fraudulent and manipulative acts and practices will be sufficient’ to justify dispensing with a surveillance-sharing agreement with a regulated market of significant size.”¹⁵³

122. Possible manipulation is an important factor U.S. regulators will consider when evaluating whether an asset is eligible to be traded on regulated exchanges. Since SOL futures may come under the purview of the CFTC, the approval of SOL-based futures may be premised on whether the CFTC believes that SOL is “not meaningfully susceptible to manipulation.” While one cannot assume that its prior approval of BTC and ETH futures translates directly to SOL, SOL shares some similarities with BTC and ETH, such as broad exchange listings, substantial market capitalization, and high and growing trading volumes. Other factors likely to be assessed by regulators include specific market attributes, including liquidity depth, token distribution, consensus structure, and market oversight.

123. We address many of these points in this paper. While we do not endeavor to provide a comprehensive review of the mechanisms through which the SOL market *could* be manipulated, we do note two compelling findings that indicate the SOL market is relatively resistant to manipulation. First, and as we describe above, trading is conducted across dozens of independent exchanges that have large volumes, deep order books, tight-spreads, and have active trading 24 hours per day. Second, SOL token prices and returns are consistent across these exchanges. The persistent high correlations suggest that the arbitrage mechanisms are working effectively. Therefore, to successfully manipulate the price of SOL on any single exchange, one would likely need to influence the global price of SOL. However, doing so would likely impose a high cost to the would-be manipulator and therefore provide a strong deterrent.

¹⁵² Securities and Exchange Commission, Release No. 34-99306, Jan 10, 2024, available at <https://www.sec.gov/files/rules/sro/nysearca/2024/34-99306.pdf>, p. 4.

¹⁵³ Securities and Exchange Commission, Release No. 34-99306, Jan 10, 2024, available at <https://www.sec.gov/files/rules/sro/nysearca/2024/34-99306.pdf>, p. 5.